

# User Manual

## Base of Aircraft Data (BADA) Family 3

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## REPORT DOCUMENTATION PAGE

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| <b>Reference</b><br>EIH Technical/Scientific Report No.<br>25/07/24-45                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Security Classification</b><br>Unclassified                                                                                                                                                                                                                                       |                                        |                     |                                 |                     |                        |
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| <b>TITLE:</b><br><div style="text-align: center; font-weight: bold;">USER MANUAL FOR THE BASE OF Aircraft DATA (BADA) FAMILY 3</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                      |                                        |                     |                                 |                     |                        |
| <b>Author</b><br>V. Mouillet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Date</b><br>09/25                                                                                                                                                                                                                                                                 | <b>Pages</b><br>xiv+61                 | <b>Figures</b><br>0 | <b>Tables</b><br>1              | <b>Annexes</b><br>1 | <b>References</b><br>7 |
| <b>Project</b><br>BADA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                      | <b>Task no. sponsor</b><br>ATD/INO/SPS |                     | <b>Period</b><br>05/22 to 09/25 |                     |                        |
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| <b>Descriptors (keywords):</b><br>aircraft model, total-energy model, BADA, user manual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                      |                                        |                     |                                 |                     |                        |
| <b>Abstract:</b><br><p>BADA (Base of Aircraft Data) is an Aircraft Performance Model (APM) applicable for aircraft trajectory simulation and prediction within the domain of Air Traffic Management (ATM).</p> <p>The APM adopted by BADA is based on a mass-varying, kinetic approach. This approach models an aircraft as a point and requires modelling of underlying forces that cause aircraft motion. The BADA APM is structured into four models: Actions, Motion, Operations and Limitations. The actions model allows the computation of the forces acting on the aircraft which cause its motion (aerodynamic, propulsive and gravitational). The propulsive model provides an associated model to compute fuel consumption. A Total Energy Model (TEM), which relates to the geometrical, kinematic and kinetic aspects of the aircraft motion, is used to calculate aircraft performances and trajectory. The operations model provides knowledge about the way the aircraft is operated, necessary to compute related aircraft motion. The limitations model restricts the aircraft behaviour in order to keep it between certain limits to ensure the safe operation of the aircraft.</p> <p>This document describes, for the BADA Family 3 APM, the mathematical models on which the data are based and the format of the files which contain the data. While these BADA model specifications are publicly accessible, the access to the BADA datasets is subject to the conclusion of a licence agreement with EUROCONTROL.</p> |                                                                                                                                                                                                                                                                                      |                                        |                     |                                 |                     |                        |

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## SUMMARY

BADA (Base of Aircraft Data) is an Aircraft Performance Model (APM) applicable for aircraft trajectory simulation and prediction within the domain of Air Traffic Management (ATM).

The APM adopted by BADA is based on a mass-varying, kinetic approach. This approach models an aircraft as a point and requires modelling of underlying forces that cause aircraft motion. The BADA APM is structured into four models: Actions, Motion, Operations and Limitations. The actions model allows the computation of the forces acting on the aircraft which cause its motion (aerodynamic, propulsive and gravitational). The propulsive model provides an associated model to compute fuel consumption. A Total Energy Model (TEM), which relates to the geometrical, kinematic and kinetic aspects of the aircraft motion, is used to calculate aircraft performances and trajectory. The operations model provides knowledge about the way the aircraft is operated, necessary to compute related aircraft motion. The limitations model restricts the aircraft behaviour in order to keep it between certain limits to ensure the safe operation of the aircraft.

This document describes, for the BADA Family 3 APM, the mathematical models on which the data are based and the format of the files which contain the data. While these BADA model specifications are publicly accessible, the access to the BADA datasets is subject to the conclusion of a licence agreement with EUROCONTROL.

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## USER MANUAL MODIFICATION HISTORY

| Issue Number | Release Date | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Version 2.1  | 31.05.94     | First release of document                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Version 2.2  | 25.01.95     | Released with BADA Revision 2.2 <ul style="list-style-type: none"> <li>- modified file formats</li> <li>- additional Synonym File</li> <li>- corrections to formulas</li> <li>- additional description of total-energy and standard atmosphere equations</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Version 2.3  | 08.06.95     | Released with BADA Revision 2.3 <ul style="list-style-type: none"> <li>- document format modified to be consistent with EEC Technical Note standards</li> <li>- maximum payload parameter added to all OPF files</li> <li>- Performance Tables Files (*.PTF) introduced</li> <li>- ISA equations used for TAS/CAS conversions instead of approximations (Section 3.2)</li> <li>- use only one formula for correction of speeds at mass values different from reference mass (Section 3.3)</li> <li>- add specification of minimum speed as function of stall speed (Section 3.4)</li> <li>- specification of transition altitude calculated added (Section 4.1)</li> <li>- speed schedules modified for climb (Section 4.1) and descent (Section 4.3)</li> <li>- modify Internet address for remote access and EUROCONTROL contact person (Section 6)</li> <li>- removed Section 7 (General Comments)</li> </ul> |
| Version 2.4  | 04.01.96     | Released with BADA Revision 2.4 <ul style="list-style-type: none"> <li>- introduction of dynamic maximum altitude</li> <li>- new temperature correction on thrust</li> <li>- ESF calculation for constant CAS below tropopause changed from binomial approximation to exact formula</li> <li>- change in altitude limit for descent speed</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Version 2.5  | 20.01.97     | <ul style="list-style-type: none"> <li>- change of minimum speeds</li> <li>- change of climb/descent speed schedules</li> <li>- cruise fuel flow correction</li> <li>- buffeting speed for jet a/c</li> <li>- addition of BADA.GPF file</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Issue Number | Release Date | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |              | <ul style="list-style-type: none"> <li>- definition of acceleration limits, bank angles and holding speeds</li> <li>- Format changes in OPF file</li> <li>- Header changes in PTF file</li> <li>- Temperature influence on thrust limitation changed</li> <li>- Unit of V<sub>stall</sub> in OPF file changed to KCAS</li> <li>- Correction of typing errors</li> <li>- Correction of APF file format explanation</li> </ul>                                                                                                                        |
| Version 2.6  | 01.09.97     | <ul style="list-style-type: none"> <li>- Change of speed schedule in the take-off / initial climb phase and approach / landing phase</li> <li>- Change in descent thrust algorithm</li> <li>- Use of exact formula for density below tropopause instead of approximation.</li> <li>- Addition of formula for pressure above tropopause</li> <li>- Change of buffeting limit to 1.2g (was 1.3g)</li> <li>- Change of OPF file format</li> <li>- Correction of some minor typing errors</li> <li>- Modified algorithm for lift coefficient</li> </ul> |
| Version 3.0  | 01.03.98     | <ul style="list-style-type: none"> <li>- Climb speed law changed for jet aircraft</li> <li>- Descent speed law changed for jet, turbo and piston</li> <li>- Reduced power climbs</li> <li>- Use of ICAO Doc. 8643/25 standard</li> <li>- Removal of C<sub>m16</sub> from drag expression</li> </ul>                                                                                                                                                                                                                                                 |
| Version 3.1  | 01.10.98     | <p>Released with BADA Revision 3.1</p> <ul style="list-style-type: none"> <li>- Removed formula for air density calculation above tropopause</li> <li>- Addition of Appendix D: Solutions for buffeting limit algorithm</li> <li>- Removed Section 3.7.2: Maximum Take-Off Thrust</li> <li>- Description for C<sub>red</sub> parameter added</li> <li>- Correction of some minor typing errors</li> <li>- Modified PTF File format (Flight Level): Section 6.6</li> <li>- Cruise CAS schedule for jet &amp; turbo aircraft (Section 4.2)</li> </ul> |
| Version 3.3  |              | <p>Released with BADA Revision 3.3</p> <ul style="list-style-type: none"> <li>- Standard atmosphere explanation added</li> <li>- Correction of some typing errors, minor changes in the layout and equations presentation.</li> </ul>                                                                                                                                                                                                                                                                                                               |
| Version 3.4  | June 2002    | <p>Released with BADA Revision 3.4</p> <ul style="list-style-type: none"> <li>- correction of some typing errors</li> <li>- in chapter 3.5 configuration threshold altitude values replaced with H<sub>max,i</sub> while the corresponding numbers are listed in chapter 5.6</li> <li>- Appendix B: a new column is added to the table; providing the information on maximum altitude that</li> </ul>                                                                                                                                               |

| Issue Number             | Release Date | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          |              | <p>an aircraft can reach at MTOW (<math>h_{max}</math>)</p> <ul style="list-style-type: none"> <li>- Modification of the value for Maximum bank angles for civil flight during HOLD in BADA.GPF file</li> <li>- Configuration Management of BADA files have been changed; files have been migrated from RCS to Continuous Configuration Management System. That resulted in the modification of the “identification” part of all BADA files given in the header.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Version 3.5              | July 2003    | <p>Released with BADA Revision 3.5</p> <ul style="list-style-type: none"> <li>- correction of some typing errors</li> <li>- SYNONYM_ALL.LST file added.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Version 3.6              | July 2004    | Released with BADA Revision 3.6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Version 3.7              | March 2009   | <p>Released with BADA Revision 3.7</p> <ul style="list-style-type: none"> <li>- Modification of the values for constants <math>g</math> and <math>R</math> in Section 3.</li> <li>- New description of formula 3.1-8 to match its actual use in some models.</li> <li>- Coefficient <math>C_{Vmin,TO}</math> is no longer used in climb speed schedule, only in flight envelope determination.</li> <li>- Numbering of several equations changed due to reorganisation of related sections.</li> <li>- Change of descent thrust computation when <math>C_{Tdes,app}</math> and <math>C_{Tdes,ld}</math> are null in Section 3.7.3.</li> <li>- Clarification of descent fuel flow computation in Section 3.9.</li> <li>- Additional information on climb and descent speed schedules in Section 4.</li> <li>- Update of some Fortran format descriptions in Section 6.</li> <li>- Additional reasons for ROCD discontinuities added in Section 6.6.</li> <li>- Introduction of new PTD file format.</li> <li>- Update of Section 7 to describe the new means of access to the BADA files.</li> </ul> |
| Version 3.8<br>Issue 1.0 | April 2010   | <p>Released with BADA Revision 3.8</p> <ul style="list-style-type: none"> <li>- Introduction of new revised atmosphere model and relevant corresponding updates throughout the User Manual document</li> <li>- Harmonisation of acronyms for physical constants with the EEC Technical Report No. 2010-001 “Revision of Atmosphere Model in BADA Aircraft Performance Model”</li> <li>- Clarification of descent fuel flow computation in Section 3.9.</li> <li>- Information added on whether some BADA model coefficients may or may not be negative.</li> <li>- Missing information about speed schedule in</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Issue Number             | Release Date | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          |              | <p>cruise for piston aircraft added (section 4.2)</p> <ul style="list-style-type: none"> <li>- Additional clarifications provided on use of altitudes in Section 4.</li> <li>- Additional explanatory note provided on data presented in the PTF file.</li> <li>- Correction of error in the solution for buffeting limit algorithm.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                              |
| Version 3.8<br>Issue 1.1 | August 2010  | <p>Clarifications only, no impact on BADA implementations:</p> <ul style="list-style-type: none"> <li>- Overall review of the document to fix formatting and typography problems.</li> <li>- Formula 3.1-19 (approximate value of a constant) removed, formula 3.1-4 added to define <math>T_{ISA,trop}</math>, and some formulas reordered in section 3.1</li> </ul>                                                                                                                                                                                                                                                                                                                                                                        |
| Version 3.9              | April 2011   | <p>Released with BADA Revision 3.9:</p> <ul style="list-style-type: none"> <li>- Minor updates in the document</li> <li>- Clarification about speed calculation in Chapter 4.2. Cruise</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Version 3.10             | April 2012   | <p>Released with BADA Revision 3.10:</p> <ul style="list-style-type: none"> <li>- Corrected Fortran specification of the PTF file to match actual release files (it would miss the first digit of descent fuel flow in some cases)</li> <li>- Clarification about the impact of speed envelope on speed calculation in Chapters 4.1, 4.2 and 4.3</li> <li>- Slight change in the description of the buffeting limit algorithm to mention that the discriminant is not “always” but “usually” negative</li> </ul>                                                                                                                                                                                                                             |
| Version 3.11             | May 2013     | <p>Released with BADA Revision 3.11:</p> <p>Special thanks to Zlata Belotic (University of Belgrade) for her contribution to this release.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Version 3.12             | August 2014  | <p>Released with BADA Revision 3.12:</p> <ul style="list-style-type: none"> <li>- Corrected Fortran specification of the Aircraft Type block in OPF file to match actual release files (section 5.4.2)</li> <li>- Addition of <math>C_{Tc,1}</math> and <math>C_{Tc,4}</math> to the list of coefficients that can be negative (section 2.11)</li> <li>- Pruning of the synonym files: only SYNONYM.NEW will now be maintained. Former SYNONYM.LST can be provided, upon request and with no support, to users of legacy systems.</li> <li>- The content of the last column of the SYNONYM.NEW file has been modified: it now indicates whether a designator is in use according to ICAO Doc 8643 (value “Y”) or not (value “N”).</li> </ul> |
| Version 3.13             | May 2015     | <p>Released with BADA Revision 3.13:</p> <ul style="list-style-type: none"> <li>- Addition of <math>C_{r1}</math> to the list of coefficients that can be negative (section 2.11)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Issue Number | Release Date   | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |                | <ul style="list-style-type: none"> <li>- The nominal bank angle for civil flight during phases other than take-off and landing has been reduced from 35° to 30° (section 4.3)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Version 3.14 | June 2017      | <p>Released with BADA Revision 3.14:</p> <ul style="list-style-type: none"> <li>- Correction of the transition altitude formula to manage transition altitudes located above the tropopause (section 2.1.2)</li> <li>- Extension of the low speed buffeting limit and associated coefficients to turboprop aircraft (section 2.6.2)</li> <li>- Update of the definition of the highest altitude used in PTF/PTD files to match actual release files (section 5.6)</li> <li>- Removal of obsolete information in section 5.2</li> <li>- Removal of section 7 “Remote file access”: information on how to access the BADA release files and documents is now provided in the BADA website User Guides.</li> </ul> |
| Version 3.15 | May 2019       | <p>Released with BADA Revision 3.15:</p> <ul style="list-style-type: none"> <li>- Clarification about the aerodynamic configuration to be used in the cruise phase (section 2.5)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Version 3.16 | May 2022       | <p>Released with BADA Revision 3.16:</p> <ul style="list-style-type: none"> <li>- Addition of electric engines (sections 2.3, 2.7, 2.9, 2.11, 3, 4.8, 4.11, 5.4.2, 5.8)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Version 3.17 | September 2025 | <p>Released with BADA Revision 3.17:</p> <ul style="list-style-type: none"> <li>- Removal of references to specific BADA datasets: deletion of former section 2 “Revision summary” and former Appendix A “List of available aircraft models”, replacement of samples files in section 5 with dummy models</li> <li>- Clarification about the fuel consumption including the contribution from all engines (section 2.9)</li> <li>- Minor clarifications and corrections throughout section 5 (file formats)</li> </ul>                                                                                                                                                                                          |

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## 1. INTRODUCTION

### 1.1. IDENTIFICATION

This document is the User Manual for the Base of Aircraft Data (BADA) Family 3.

### 1.2. PURPOSE

BADA (Base of Aircraft DATA) is an Aircraft Performance Model (APM) applicable for aircraft trajectory simulation and prediction within the domain of Air Traffic Management (ATM).

The APM adopted by BADA is based on a mass-varying, kinetic approach. This approach models an aircraft as a point and requires modelling of underlying forces that cause aircraft motion. The BADA APM is structured into four models: Actions, Motion, Operations and Limitations. The actions model allows the computation of the forces acting on the aircraft which cause its motion (aerodynamic, propulsive and gravitational). The propulsive model provides an associated model to compute fuel consumption. A Total Energy Model (TEM), which relates to the geometrical, kinematic and kinetic aspects of the aircraft motion, is used to calculate aircraft performances and trajectory. The operations model provides knowledge about the way the aircraft is operated, necessary to compute related aircraft motion. The limitations model restricts the aircraft behaviour in order to keep it between certain limits to ensure the safe operation of the aircraft.

This document describes, for the BADA Family 3 APM, the mathematical models on which the data are based and the format of the files which contain the data. While these BADA model specifications are publicly accessible<sup>1</sup>, the access to the BADA datasets is subject to the conclusion of a licence agreement with EUROCONTROL<sup>2</sup>.

### 1.3. DOCUMENT ORGANISATION

This document consists of five sections including Section 1, the Introduction. A list of referenced documents along with a glossary of acronyms and symbols are included in this section.

**Section 2:** Operation Performance Models, defines the set of equations that are used to parameterise aircraft performance. This includes models of aerodynamic drag, engine thrust, and fuel consumption. An atmosphere model is also provided.

**Section 3:** Airline Procedure Models, defines the set of parameters that is used to characterise standard airline speed procedures for climb, cruise, and descent.

**Section 4:** Global Aircraft Parameters, defines the set of global aircraft parameters that are valid for all, or a group of, aircraft.

**Section 5:** File Structure, describes the files in which the BADA aircraft parameters are maintained.

One appendix is also provided with this document. **Appendix A** gives solutions for a buffeting limit algorithm.

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<sup>1</sup> <https://github.com/eurocontrol-bada>

<sup>2</sup> <https://www.eurocontrol.int/model/bada>

#### 1.4. REFERENCED DOCUMENTS

- RD1** Aircraft Type Designators; ICAO Document No. 8643; Edition 53, April 2025, <http://www.icao.int/publications/DOC8643/>
- RD2** Aircraft Modelling Standards for Future ATC Systems; EUROCONTROL Division E1 Document No. 872003, July 1987.
- RD3** Manual of the ICAO Standard Atmosphere; ICAO Document No. 7488, 2nd Edition, 1964.
- RD4** Base of Aircraft Data (BADA) Aircraft Performance Modelling Manual; EEC Technical Report No. 2009-009, April 2009.
- RD5** Memo on the Calculation of Energy Share Factor; EEC/FAS/BYR/95/50; 22 November 1995.
- RD6** Revision of Atmosphere Model in BADA Aircraft Performance Model; EEC Technical Report No. 2010-001, February 2010.
- RD7** Mathematical Handbook; M.R. Spiegel; 1968; McGraw-Hill book company.

## 1.5. GLOSSARY OF ACRONYMS

|              |                                                                                                   |
|--------------|---------------------------------------------------------------------------------------------------|
| <b>AGL</b>   | <b>A</b> bove <b>G</b> round <b>L</b> evel                                                        |
| <b>APF</b>   | <b>A</b> irlines <b>P</b> rocedures <b>F</b> ile                                                  |
| <b>ASCII</b> | <b>A</b> merican <b>S</b> tandard <b>C</b> ode for the <b>I</b> nterchange of <b>I</b> nformation |
| <b>ATM</b>   | <b>A</b> ir <b>T</b> raffic <b>M</b> anagement                                                    |
| <b>BADA</b>  | <b>B</b> ase of <b>A</b> ircraft <b>D</b> ata                                                     |
| <b>CAS</b>   | <b>C</b> alibrated <b>A</b> irspeed                                                               |
| <b>EIH</b>   | <b>E</b> UROCONTROL <b>I</b> nnovation <b>H</b> ub                                                |
| <b>ESF</b>   | <b>E</b> nergy <b>S</b> hare <b>F</b> actor                                                       |
| <b>ICAO</b>  | <b>I</b> nternational <b>C</b> ivil <b>A</b> viation <b>O</b> rganisation                         |
| <b>ISA</b>   | <b>I</b> nternational <b>S</b> tandard <b>A</b> tmosphere                                         |
| <b>MLW</b>   | <b>M</b> aximum <b>L</b> anding <b>W</b> eight                                                    |
| <b>MSL</b>   | <b>M</b> ean <b>S</b> ea <b>L</b> evel                                                            |
| <b>MTOW</b>  | <b>M</b> aximum <b>T</b> ake-off <b>W</b> eight                                                   |
| <b>OPF</b>   | <b>O</b> perations <b>P</b> erformance <b>F</b> ile                                               |
| <b>PTD</b>   | <b>P</b> erformance <b>T</b> able <b>D</b> ata                                                    |
| <b>PTF</b>   | <b>P</b> erformance <b>T</b> able <b>F</b> ile                                                    |
| <b>ROCD</b>  | <b>R</b> ate of <b>C</b> limb or <b>D</b> escent                                                  |
| <b>TAS</b>   | <b>T</b> rue <b>A</b> irspeed                                                                     |
| <b>TBP</b>   | <b>T</b> urboprop                                                                                 |
| <b>TEM</b>   | <b>T</b> otal- <b>E</b> nergy <b>M</b> odel                                                       |

## 1.6. GLOSSARY OF SYMBOLS

A list of the symbols used in equations throughout this document is given below along with a description. Where appropriate, the engineering units typically associated with the symbol are also given.

|                 |                                |                                       |
|-----------------|--------------------------------|---------------------------------------|
| $a$             | speed of sound                 | [m/s]                                 |
| $d$             | distance                       | [nautical miles]                      |
| $f$             | fuel flow                      | [kg/min]                              |
| $g_0$           | gravitational acceleration     | [m/s <sup>2</sup> ]                   |
| $\frac{dh}{dt}$ | vertical speed                 | [m/s] or [ft/min]                     |
| $h$             | geodetic altitude              | [metres] or [ft]                      |
| $H$             | geopotential altitude          | [metres] or [ft]                      |
| $H_p$           | geopotential pressure altitude | [metres] or [ft]                      |
| $C$             | general coefficient            |                                       |
| $D$             | drag force                     | [Newtons]                             |
| $m$             | aircraft mass                  | [tonnes] or [kg]                      |
| $M$             | Mach number                    | [-]                                   |
| $p$             | Actual pressure                | [Pa]                                  |
| $p_0$           | Standard pressure at MSL       | [Pa]                                  |
| $R$             | real gas constant for air      | [m <sup>2</sup> /(K·s <sup>2</sup> )] |
| ROCD            | Rate of Climb or Descent       | [m/s] or [ft/min]                     |
| $S$             | reference wing surface area    | [m <sup>2</sup> ]                     |
| $T$             | temperature                    | [Kelvin]                              |
| $Thr$           | thrust                         | [N]                                   |
| $V$             | speed                          | [m/s] or [knots]                      |
| $\Delta T$      | temperature difference         | [Kelvin]                              |
| $W$             | weight                         | [N]                                   |
| $\eta$          | thrust specific fuel flow      | [kg/(min·kN)]                         |
| $\rho$          | air density                    | [kg/m <sup>3</sup> ]                  |

## 2. OPERATIONS PERFORMANCE MODEL

This section defines the various equations and coefficients used by the BADA operations performance model.

The first two subsections describe the equations for atmospheric properties and the Total-Energy Model (TEM) equations respectively.

The remaining eight subsections define the aircraft model in terms of the eight categories listed below:

- aircraft type,
- mass,
- flight envelope,
- aerodynamics,
- engine thrust,
- reduced power,
- fuel consumption,
- ground movement.

### 2.1. ATMOSPHERE MODEL

This section provides expressions for the atmospheric properties (pressure, temperature, density and speed of sound) as a function of altitude which are required for calculation of aircraft performances and movements<sup>3</sup>. Conversions from CAS to TAS and Mach number also require the determination of several atmospheric properties as a function of altitude.

The most important equations for atmospheric properties used by BADA and CAS/TAS conversion are summarised in this chapter, while other expressions and more details are provided in [RD6].

#### 2.1.1. Definitions

Mean Sea Level (MSL) Standard atmosphere conditions are those that occur in the International Standard Atmosphere (ISA) at the point where the geopotential pressure altitude  $H_p$ <sup>4</sup> is zero. They are denoted as  $T_0$ ,  $p_0$ ,  $\rho_0$  and  $a_0$  with the values listed below:

|                                          |          |   |         |                      |
|------------------------------------------|----------|---|---------|----------------------|
| Standard atmospheric temperature at MSL: | $T_0$    | = | 288.15  | [K]                  |
| Standard atmospheric pressure at MSL:    | $p_0$    | = | 101325  | [Pa]                 |
| Standard atmospheric density at MSL:     | $\rho_0$ | = | 1.225   | [kg/m <sup>3</sup> ] |
| Speed of sound:                          | $a_0$    | = | 340.294 | [m/s]                |

<sup>3</sup> These equations are based on the International Standard Atmosphere (ISA) [RD3].

<sup>4</sup> Geopotential pressure altitude  $H_p$  is the geopotential altitude  $H$  that occurs in the ISA atmospheric conditions [RD6].

Mean Sea Level (MSL) atmosphere conditions are those that occur in a non-ISA atmosphere. They are identified by the sub-index MSL and differ from  $(T_0, p_0, \rho_0, a_0)$  in non-ISA conditions.

Non-ISA atmospheres are those that follow the same hypotheses as the ISA atmosphere but differ from it in that one or both of the following parameters is not zero:

1.  **$\Delta T$** . Temperature differential at MSL. It is the difference in atmospheric temperature at MSL between a given non-standard atmosphere and ISA.
2.  **$\Delta p$** . Pressure differential at MSL. It is the difference in atmospheric pressure at MSL between a given non-standard atmosphere and ISA.

The values of these two parameters uniquely identify any non-ISA atmosphere. Thus, a non-ISA atmosphere provides expressions for the atmospheric pressure, temperature and density as functions of the geopotential altitude  $H^5$  and its two differentials. [RD6] provides more details on the corresponding analytical expressions.

### 2.1.2. Expressions

The relationships linking the atmospheric pressure  $p$ , temperature  $T$ , geopotential pressure altitude  $H_p$  and geopotential altitude  $H$  for any ISA<sup>6</sup> and non-ISA atmosphere are provided below.

Physical constants which are used throughout this chapter are listed below:

$$\text{Adiabatic index of air: } \kappa = 1.4$$

$$\text{Real gas constant for air: } R = 287.05287 \text{ [m}^2\text{/(K}\cdot\text{s}^2\text{)]}$$

$$\text{Gravitational acceleration: } g_0 = 9.80665 \text{ [m/s}^2\text{]}$$

$$\begin{array}{l} \text{ISA temperature gradient with altitude} \\ \text{below the tropopause: } \beta_{T,<} = -0.0065 \text{ [K/m]} \end{array}$$

Note that subindex < denotes values below and at the tropopause and subindex > denotes values above the tropopause (as defined by 2.1-11).

Standard Mean Sea Level (subindex  $H_p = 0$ )

The temperature differential  $\Delta T$  sets the value of the real temperature  $T$  in non-standard atmospheres.

$$H_{p,H_p=0} = 0 \tag{2.1-1}$$

<sup>5</sup> Geopotential altitude  $H$  is that which under the standard constant gravitational field provides the same differential work performed by the standard acceleration of free fall when displacing the unit of mass a distance  $dH$  along the line of force, as that performed by the geopotential acceleration when displacing the unit of mass a geodetic distance  $dh$  [RD6].

<sup>6</sup> By replacing  $\Delta T$  and  $\Delta p$  parameters with zeros the expressions are made applicable to the case of the standard atmosphere.

$$p_{Hp=0} = p_0 \quad (2.1-2)$$

$$T_{ISA,Hp=0} = T_0 \quad (2.1-3)$$

$$T_{Hp=0} = T_0 + \Delta T \quad (2.1-4)$$

$$H_{Hp=0} = \frac{1}{\beta_{T,<}} \left[ T_0 - T_{ISA,MSL} + \Delta T \cdot \ln \left( \frac{T_0}{T_{ISA,MSL}} \right) \right] \quad (2.1-5)$$

where  $T_{ISA}$  is the standard atmospheric temperature that occurs in the ISA atmosphere. It is a function of the geopotential pressure altitude  $H_p$ .

### Mean Sea Level (subindex MSL)

The pressure differential  $\Delta p$  sets the value of the atmospheric pressure  $p$ .

$$H_{MSL}=0 \quad (2.1-6)^7$$

$$p_{MSL} = p_0 + \Delta p \quad (2.1-7)$$

$$H_{p,MSL} = \frac{T_0}{\beta_{T,<}} \left[ \left( \frac{p_{MSL}}{p_0} \right)^{\frac{\beta_{T,<} R}{g_0}} - 1 \right] \quad (2.1-8)$$

$$T_{ISA,MSL} = T_0 + \beta_{T,<} H_{p,MSL} \quad (2.1-9)$$

$$T_{MSL} = T_0 + \Delta T + \beta_{T,<} H_{p,MSL} \quad (2.1-10)$$

### Tropopause

Tropopause is the separation between two different layers: the troposphere, which stands below it, and the stratosphere, which is placed above. Its altitude  $H_{p,trop}$  is constant when expressed in terms of geopotential pressure altitude:

$$H_{p,trop} = 11000 \text{ [m]} \quad (2.1-11)$$

<sup>7</sup> In order to simplify the expressions, this document assumes that the geopotential altitude at mean sea level is always zero.

a) Determination of Temperature

$$T = f(H_p, \Delta T) \quad (2.1-12)$$

$$T_{<} = T_0 + \Delta T + \beta_{T,<} H_{p,<} \quad (2.1-13)$$

$$T_{ISA,trop} = T_0 + \beta_{T,<} H_{p,trop} \quad (2.1-14)$$

$$T_{trop} = T_0 + \Delta T + \beta_{T,<} H_{p,trop} \quad (2.1-15)$$

$$T_{>} = T_{trop} \quad (2.1-16)$$

b) Determination of Air Pressure

$$p = f(T, \Delta T) \quad (2.1-17)$$

$$p_{<} = p_0 \left( \frac{T_{<} - \Delta T}{T_0} \right)^{-\frac{g_0}{\beta_{T,<} R}} \quad (2.1-18)$$

$$p_{trop} = p_0 \left( \frac{T_{trop} - \Delta T}{T_0} \right)^{-\frac{g_0}{\beta_{T,<} R}} \quad (2.1-19)$$

$T_{>} = T_{trop}$ , so  $p_{>}$  does not directly depend on temperature  $T_{>}$ . For altitudes above the tropopause, the following formula should be used:

$$p_{>} = p_{trop} \exp \left[ -\frac{g_0}{R T_{ISA,trop}} (H_{p,>} - H_{p,trop}) \right] \quad (2.1-20)$$

where altitudes  $H_{p,>}$  and  $H_{p,trop}$  are expressed in metres.

c) Determination of Air Density

The air density,  $\rho$ , in  $\text{kg/m}^3$ , is calculated from the pressure  $p$  and the temperature  $T$  at altitude using the perfect gas law:

$$\rho = \frac{p}{RT} \quad (2.1-21)$$

d) Determination of Speed of Sound

The speed of sound,  $a$ , is the speed at which the pressure waves travel through a fluid and it is given by the expression:

$$a = \sqrt{\kappa R T} \quad (2.1-22)$$

e) CAS/TAS Conversion

The true airspeed,  $V_{TAS}$  [m/s], is calculated as a function of the calibrated air speed,  $V_{CAS}$  [m/s], as follows:

$$V_{TAS} = \left[ \frac{2 p}{\mu \rho} \left\{ 1 + \frac{p_0}{p} \left[ \left( 1 + \frac{\mu \rho_0}{2 p_0} V_{CAS}^2 \right)^{\frac{1}{\mu}} - 1 \right]^\mu - 1 \right\} \right]^{\frac{1}{2}} \quad (2.1-23)$$

Similarly,  $V_{CAS}$  [m/s] is calculated as a function of  $V_{TAS}$  [m/s] as follows:

$$V_{CAS} = \left[ \frac{2 p_0}{\mu \rho_0} \left\{ 1 + \frac{p}{p_0} \left[ \left( 1 + \frac{\mu \rho}{2 p} V_{TAS}^2 \right)^{\frac{1}{\mu}} - 1 \right]^\mu - 1 \right\} \right]^{\frac{1}{2}} \quad (2.1-24)$$

where symbols not previously defined are explained below:

$$\mu = \frac{\kappa - 1}{\kappa} \quad \left( \mu = \frac{1}{3.5} \text{ if } \kappa = 1.4 \right) \quad (2.1-25)$$

f) Mach/TAS conversion

The true airspeed,  $V_{TAS}$  [m/s], is calculated as a function of the Mach number,  $M$ , as follows:

$$V_{TAS} = M \times \sqrt{\kappa R T} \quad (2.1-26)$$

g) Mach/CAS transition altitude

The transition altitude (also called crossover altitude),  $H_{p,trans}$  [m], between a given CAS,  $V_{CAS}$  [m/s], and a Mach number,  $M$ , is defined to be the geopotential pressure altitude at which  $V_{CAS}$  and  $M$  represent the same TAS value, and can be calculated as follows:

$$H_{p,trans} = \begin{cases} \frac{T_0}{\beta_{T,<}} \cdot \left[ \left( \frac{p_{trans}}{p_0} \right)^{\frac{\beta_{T,<} R}{g_0}} - 1 \right] & \text{when } p_{trans} \geq p_{trop} \\ H_{P,trop} - \frac{R \cdot T_{ISA,trop}}{g_0} \cdot \ln \left( \frac{p_{trans}}{p_{trop}} \right) & \text{when } p_{trans} < p_{trop} \end{cases} \quad (2.1-27)$$

where  $p_{trans}$  is the pressure at the transition altitude,

$$p_{trans} = p_0 \cdot \frac{\left[ 1 + \left( \frac{\kappa - 1}{2} \right) \left( \frac{V_{CAS}}{a_0} \right)^2 \right]^{\frac{\kappa}{\kappa - 1}} - 1}{\left[ 1 + \frac{\kappa - 1}{2} M^2 \right]^{\frac{\kappa}{\kappa - 1}} - 1} \quad (2.1-28)$$

## 2.2. TOTAL-ENERGY MODEL

The Total-Energy Model equates the rate of work done by forces acting on the aircraft to the rate of increase in potential and kinetic energy, that is:

$$(Thr - D) \cdot V_{TAS} = mg_0 \frac{dh}{dt} + mV_{TAS} \frac{dV_{TAS}}{dt} \quad (2.2-1)$$

The symbols are defined below with metric units specified:

|                  |   |                                                        |                             |
|------------------|---|--------------------------------------------------------|-----------------------------|
| Thr              | - | thrust acting parallel to the aircraft velocity vector | [Newtons]                   |
| D                | - | aerodynamic drag                                       | [Newtons]                   |
| m                | - | aircraft mass                                          | [kilograms]                 |
| h                | - | geodetic altitude                                      | [m]                         |
| g <sub>0</sub>   | - | gravitational acceleration                             | [9.80665 m/s <sup>2</sup> ] |
| V <sub>TAS</sub> | - | true airspeed                                          | [m/s]                       |
| $\frac{d}{dt}$   | - | time derivative                                        | [s <sup>-1</sup> ]          |

Note that true airspeed is often calculated in knots and altitude calculated in feet thus requiring the appropriate conversion factors.

Without considering the use of devices such as spoilers, leading-edge slats or trailing-edge flaps, there are two independent control inputs available for affecting the aircraft trajectory in the vertical plane. These are the throttle and the elevator.

These inputs allow any two of the three variables of thrust, speed, or rate of climb or descent (ROCD) to be controlled. The other variable is then determined by equation 2.2-1. The three resulting control possibilities are elaborated on below.

(a) Speed and Throttle Controlled - Calculation of Rate of Climb or Descent

Assuming that velocity and thrust are independently controlled, then equation 2.2-1 is used to calculate the resulting rate of climb or descent (ROCD). This is a fairly common case for climbs and descents in which the throttle is set to some fixed position (maximum climb thrust or idle for descent) and the speed is maintained at some constant value of calibrated airspeed (CAS) or Mach number.

(b) ROCD and Throttle Controlled - Calculation of Speed

Assuming that the ROCD and thrust are independently controlled, then equation 2.2-1 is used to calculate the resulting speed.

(c) Speed and ROCD Controlled - Calculation of Thrust

Assuming that both ROCD and speed are controlled, then equation 2.2-1 can be used to calculate the necessary thrust. This thrust must be within the available limits for the desired ROCD and speed to be maintained.

Case (a), above, is the most common such that equation 2.2-1 is most often used to calculate the rate of climb or descent. To facilitate this calculation, equation 2.2-1 can be rearranged as follows:

$$(Thr - D) \cdot V_{TAS} = mg_0 \frac{dh}{dt} + m V_{TAS} \left( \frac{dV_{TAS}}{dh} \right) \left( \frac{dh}{dt} \right) \quad (2.2-2)$$

Isolating the vertical speed on the left-hand side gives:

$$\frac{dh}{dt} = \frac{(Thr - D) \cdot V_{TAS}}{mg_0} \left[ 1 + \left( \frac{V_{TAS}}{g_0} \right) \left( \frac{dV_{TAS}}{dh} \right) \right]^{-1} \quad (2.2-3)$$

Vertical speed is defined as the variation with time of the aircraft geodetic altitude  $h$ . The assumption of a standard constant gravity field derives in identical geodetic and geopotential altitudes  $H$  [RD6].

The ROCD is defined as the variation with time of the aircraft geopotential pressure altitude  $H_p$ . It is the preferred way of presenting the performances of an aircraft as it eliminates possible variations caused by the atmospheric conditions:

$$ROCD = \frac{dH_p}{dt} = \frac{T - \Delta T}{T} \frac{(Thr - D) \cdot V_{TAS}}{mg_0} \left[ 1 + \left( \frac{V_{TAS}}{g_0} \right) \left( \frac{dV_{TAS}}{dh} \right) \right]^{-1} \quad (2.2-4)$$

where:

- $T$  - atmosphere temperature [K];
- $\Delta T$  - temperature differential [K].

It has been shown by Renteux [RD2] that the last term can be replaced by an energy share factor as a function of Mach number,  $f\{M\}$ :

$$f\{M\} = \left[ 1 + \left( \frac{V_{TAS}}{g_0} \right) \cdot \left( \frac{dV_{TAS}}{dh} \right) \right]^{-1} \quad (2.2-5)$$

This leads to:

$$\frac{dh}{dt} = \left[ \frac{(Thr - D) \cdot V_{TAS}}{mg_0} \right] f\{M\} \quad (2.2-6)$$

$$ROCD = \frac{dH_p}{dt} = \frac{T - \Delta T}{T} \left[ \frac{(Thr - D) \cdot V_{TAS}}{mg_0} \right] f\{M\} \quad (2.2-7)$$

This energy share factor  $f\{M\}$  specifies how much of the available power is allocated to climb as opposed to acceleration while following a selected speed profile during climb.

For several common flight conditions, equation 2.2-5 can be rewritten as is done below [RD5]:

- (a) Constant Mach number in stratosphere (i.e. above tropopause)

$$f\{M\} = 1.0 \quad (2.2-8)$$

Note that above the tropopause the air temperature and the speed of sound are constant. Maintaining a constant Mach number therefore requires no acceleration and all available power can be allocated to a change in altitude.

- (b) Constant Mach number below tropopause:

$$f\{M\} = \left[ 1 + \frac{\kappa R \beta_{T,<}}{2 g_0} M^2 \frac{T - \Delta T}{T} \right]^{-1} \quad (2.2-9)$$

In this case, for a typical Mach number of 0.8 the energy share factor allocated to climb is 1.09.

This number is greater than 1 because below the tropopause, the temperature and thus, speed of sound decreases with altitude. Maintaining a constant Mach number during climb thus means that the true airspeed decreases with altitude. Consequently, the rate of climb benefits from not only all the available power but also a transfer of kinetic energy to potential energy.

- (c) Constant Calibrated Airspeed (CAS) below tropopause

$$f\{M\} = \left\{ 1 + \frac{\kappa R \beta_{T,<}}{2 g_0} M^2 \frac{T - \Delta T}{T} + \left( 1 + \frac{\kappa - 1}{2} M^2 \right)^{\frac{-1}{\kappa - 1}} \left\{ \left( 1 + \frac{\kappa - 1}{2} M^2 \right)^{\frac{\kappa}{\kappa - 1}} - 1 \right\} \right\}^{-1} \quad (2.2-10)$$

In this case the energy share factor is less than one. A Mach number of 0.6 for example yields an energy share factor of 0.85.

This number is less than 1 because as density decreases with altitude, maintaining a constant CAS during climb requires maintaining a continual increase in true airspeed. Thus, some of the available power needs to be allocated to acceleration leaving the remainder for climb.

- (d) Constant Calibrated Airspeed (CAS) above tropopause.

$$f\{M\} = \left\{ 1 + \left( 1 + \frac{\kappa - 1}{2} M^2 \right)^{\frac{-1}{\kappa - 1}} \left\{ \left( 1 + \frac{\kappa - 1}{2} M^2 \right)^{\frac{\kappa}{\kappa - 1}} - 1 \right\} \right\}^{-1} \quad (2.2-11)$$

This formula is identical to (2.2-10), except that  $\beta_T$  is now null since we are above the tropopause.

The energy share factors given above apply equally well to descent as to climb. The difference being that the available power is negative for descent.

In cases where neither constant Mach number nor constant CAS is maintained, the following energy share factors are used:

- acceleration in climb:  $f\{M\} = 0.3$
- deceleration in descent:  $f\{M\} = 0.3$
- deceleration in climb:  $f\{M\} = 1.7$
- acceleration in descent:  $f\{M\} = 1.7$

Note that, for the cases of acceleration in climb or deceleration in descent, the majority of the available power is devoted to a change in speed.

For the cases of deceleration in climb or acceleration in descent, the energy share factor is greater than 1 since the change of altitude benefits from a transfer of kinetic energy.

## 2.3. AIRCRAFT TYPE

Three values are specified for aircraft type, these being the number of engines,  $n_{eng}$ , the engine type and the wake category.

The engine type can be one of four values:

- Jet
- Turboprop
- Piston
- Electric

The wake category can be one of four values:

- J : jumbo
- H: heavy
- M: medium
- L : light

Note that ICAO associates a wake category with each aircraft type designator [RD1].

## 2.4. MASS

Four mass values are specified for each aircraft in tonnes:

- $m_{min}$  - minimum mass
- $m_{max}$  - maximum mass
- $m_{ref}$  - reference mass
- $m_{pyld}$  - maximum payload mass

Note that the specified mass limits are taken from aircraft performance reference data which is available in the BADA library. In function of specific aircraft certified limitations, a particular aircraft version of a given aircraft type (model) may have different limits. More details on the way the mass limits are selected in BADA are provided in [RD4].

Aircraft operating speeds vary with the aircraft mass. This variation is calculated according to the formula below:

$$V = V_{ref} \times \sqrt{\frac{m}{m_{ref}}} \quad (2.4-1)$$

In this formula, the aircraft reference speed  $V_{ref}$  is given for the reference mass  $m_{ref}$ . The speed at another mass,  $m$ , is then calculated as  $V$ .

An example of an aircraft speed which can be calculated via this formula is the stall speed,  $V_{stall}$ .

## 2.5. FLIGHT ENVELOPE

### (a) Maximum Speed and Altitude

The maximum speed and altitude for an aircraft are expressed in terms of the following six parameters:

|           |   |                                                                                                                             |
|-----------|---|-----------------------------------------------------------------------------------------------------------------------------|
| $V_{MO}$  | - | maximum operating speed (CAS) [kt]                                                                                          |
| $M_{MO}$  | - | maximum operational Mach number                                                                                             |
| $h_{MO}$  | - | maximum operating altitude [ft] above standard MSL                                                                          |
| $h_{max}$ | - | maximum altitude [ft] above standard MSL at MTOW under ISA conditions (allowing about 300 ft/min of residual rate of climb) |
| $G_w$     | - | mass gradient on $h_{max}$ [ft/kg]                                                                                          |
| $G_t$     | - | temperature gradient on $h_{max}$ [ft/K]                                                                                    |

The maximum altitude for any given mass is:

$$h_{max/act} = \text{MIN} [h_{MO}, h_{max} + G_t \times (\Delta T - C_{Tc,4}) + G_w \times (m_{max} - m_{act})] \quad (2.5-1)$$

where:  $\Delta T$  is the temperature deviation from ISA [K]  
 $m_{act}$  is the actual aircraft mass [kg]

with:  $G_w \geq 0$

$G_t \leq 0$

if  $(\Delta T - C_{Tc,4}) < 0$ , then:  $(\Delta T - C_{Tc,4}) = 0$

Formula 2.5-1 should not be executed when the  $h_{max}$  value in the OPF file is set to 0 (zero). In that case the maximum altitude is always  $h_{MO}$ .

Note that the given speed and altitude limits are taken from available reference data: depending upon specific certifications, a particular aircraft of a given type may present different limits.

### (b) Minimum Speed

The minimum speed for the aircraft is in function of aircraft stall speed and specified as follows:

$$V_{min} = C_{vmin,TO} \times V_{stall} \quad \text{if in take-off} \quad (2.5-2)$$

$$V_{min} = C_{vmin} \times V_{stall} \quad \text{otherwise} \quad (2.5-3)$$

Note: See Section 2.6.2 for minimum speed at high altitude for jet aircraft and Section 4.7 for the values of the minimum speed coefficients.

Here the speeds are specified in terms of CAS. The stall speed depends upon the configuration.

Specifically, five different configurations are specified with a stall speed,  $(V_{\text{stall}})_i$ , and configuration threshold altitude,  $H_{\text{max},i}$ , given for each:

|                                                                                                                                                                                                                                                                                                                                                       |                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| TO - take-off configuration<br>(in climb up to $H_{\text{max},\text{TO}}$ AGL)                                                                                                                                                                                                                                                                        | $(V_{\text{stall}})_{\text{TO}}$ |
| IC - initial climb configuration<br>(in climb between $H_{\text{max},\text{TO}}$ and $H_{\text{max},\text{IC}}$ AGL)                                                                                                                                                                                                                                  | $(V_{\text{stall}})_{\text{IC}}$ |
| CR - cruise (clean) configuration<br>(in climb above $H_{\text{max},\text{IC}}$ AGL,<br>in cruise,<br>in descent above $H_{\text{max},\text{AP}}$ AGL,<br>in descent below $H_{\text{max},\text{AP}}$ AGL when<br>$V \geq V_{\text{min},\text{cruise}} + 10 \text{ kt}$ )                                                                             | $(V_{\text{stall}})_{\text{CR}}$ |
| AP - approach configuration<br>(in descent between $H_{\text{max},\text{AP}}$ AGL and $H_{\text{max},\text{LD}}$ AGL when<br>$V < V_{\text{min},\text{cruise}} + 10 \text{ kt}$ ,<br>in descent below $H_{\text{max},\text{LD}}$ AGL when<br>$V_{\text{min},\text{cruise}} + 10 \text{ kt} > V \geq V_{\text{min},\text{approach}} + 10 \text{ kt}$ ) | $(V_{\text{stall}})_{\text{AP}}$ |
| LD - landing configuration<br>(in descent below $H_{\text{max},\text{LD}}$ AGL when<br>$V < V_{\text{min},\text{approach}} + 10 \text{ kt}$ )                                                                                                                                                                                                         | $(V_{\text{stall}})_{\text{LD}}$ |

The threshold altitudes are expressed in terms of geopotential pressure altitude. However, when aircraft operations close to the ground are considered, one has to account for airport/runway elevation<sup>8</sup>. The pressure altitude thresholds provided above correspond to geopotential pressure altitude Above Ground Level (AGL).

The stall speeds correspond to a minimum stall speed and not a 1-g stall speed. Also, the BADA model assumes that for any aircraft these stall speeds have the following relationship:

$$(V_{\text{stall}})_{\text{CR}} \geq (V_{\text{stall}})_{\text{IC}} \geq (V_{\text{stall}})_{\text{TO}} \geq (V_{\text{stall}})_{\text{AP}} \geq (V_{\text{stall}})_{\text{LD}}$$

The configuration specific values are listed in Section 4.6. The speeds  $V$  used during the descent, approach and landing phases are defined in Section 3.3.

---

<sup>8</sup> Measured from Mean Sea Level (MSL).

## 2.6. AERODYNAMICS

### 2.6.1. Aerodynamic Drag

The lift coefficient,  $C_L$ , is determined assuming that the flight path angle is zero. However, a correction for a bank angle  $\phi$  is made.

$$C_L = \frac{2 \cdot m \cdot g_0}{\rho \cdot V_{TAS}^2 \cdot S \cdot \cos \phi} \quad (2.6-1)$$

Under nominal conditions, the drag coefficient,  $C_D$  is specified as a function of the lift coefficient  $C_L$  as follows:

$$C_D = C_{D0,CR} + C_{D2,CR} \times (C_L)^2 \quad (2.6-2)$$

Formula 2.6-2 is valid for all situations except for the approach and landing where other drag coefficients are to be used.

In the approach configuration (as defined in Section 2.5) a different flap setting is used, and formula 2.6-3 should be applied:

$$C_D = C_{D0,AP} + C_{D2,AP} \times (C_L)^2 \quad (2.6-3)$$

In the landing configuration (as defined in Section 2.5) a different flap setting is used, and formula 2.6-4 should be applied:

$$C_D = C_{D0,LDG} + C_{D0,\Delta LDG} + C_{D2,LDG} \times (C_L)^2 \quad (2.6-4)$$

The value of  $C_{D0,\Delta LDG}$  represents drag increase due to the landing gear. The values of  $C_{D0,LD}$  in the OPF files were all determined for the landing flap setting mentioned in the OPF file.

The drag force [Newtons] is then determined from the drag coefficient in the standard manner:

$$D = \frac{C_D \cdot \rho \cdot V_{TAS}^2 \cdot S}{2} \quad (2.6-5)$$

Where:

$\rho$  is the air density [kg/m<sup>3</sup>]

$S$  is the wing reference area [m<sup>2</sup>]

$V_{TAS}$  is the true airspeed [m/s].

Note that the air density is a function of altitude as described in Section 2.1.

The above equations thus result in eight coefficients for the specification of drag:

$$\begin{array}{l} S \\ C_{D0,CR} \quad C_{D2,CR} \\ C_{D0,AP} \quad C_{D2,AP} \\ C_{D0,LD} \quad C_{D2,LD} \\ C_{D0,\Delta LDG} \end{array}$$

In case the  $C_{D0,AP}$ ,  $C_{D2,AP}$ ,  $C_{D0,LD}$ ,  $C_{D2,LD}$  and  $C_{D0,\Delta LDG}$  coefficients (referred to as “non-clean” data in this document) are set to 0 (zero) in the OPF file, expression 2.6-2 will be used in all cases.

## 2.6.2. Low Speed Buffeting Limit

For jet and turboprop aircraft, a low-speed buffeting limit has been introduced. This buffeting limit is expressed as a Mach number and can be determined using the following equation:

$$k \times M^3 - C_{Lbo (M=0)} \times M^2 + \frac{W}{S \cdot p \cdot 0.583} = 0 \quad (2.6-6)$$

where:

$k$  is the lift coefficient gradient

$C_{Lbo (M=0)}$  is the initial buffet onset lift coefficient for  $M=0$

$p$  is the actual pressure [Pa]

$M$  is the Mach number

$S$  is the wing reference area [m<sup>2</sup>]

$W$  is the aircraft weight [N]

Note that the factor of 0.583 gives a 0.2 g margin.

The  $k$  and  $C_{Lbo (M=0)}$  parameters have been determined for many jet aircraft and one turboprop aircraft. If the  $k$  and  $C_{Lbo (M=0)}$  parameters in the OPF file are set to 0 (zero), the minimum speed is given by expressions 2.5-2 and 2.5-3. Otherwise, the solution for  $M$  in formula 2.6-6 can be obtained using the method given in Appendix A. The buffeting limit should be applied as a minimum speed in the following way:

- If ( $H_p \geq 15,000$  ft) then:  $V_{min} = \text{MAX}(V_{min,stall}, M_b)$

- If ( $H_p < 15,000$  ft) then:  $V_{min} = V_{min,stall}$

where:  $H_p$  is the geopotential pressure altitude

$M_b$  is the lowest positive solution of expression 2.6-6

$V_{min,stall}$  is given by expressions 2.5-2 and 2.5-3

Note that the units of the two values  $V_{min,stall}$  and  $M_b$  inside the  $\text{MAX}()$  expression should be the same.

## 2.7. ENGINE THRUST

The BADA model provides coefficients that allow the calculation of the following thrust levels:

- maximum climb and take-off,
- maximum cruise,
- descent.

The thrust is calculated in Newtons and includes the contribution from all engines. The subsections below provide the equations for each of the thrust conditions.

### 2.7.1. Maximum Climb and Take-Off Thrust

The maximum climb thrust at standard atmosphere conditions,  $(Thr_{\max \text{ climb}})_{ISA}$ , is calculated in Newtons as a function of the following information:

- engine type: either Jet, Turboprop, Piston or Electric;
- geopotential pressure altitude,  $H_p$  [ft];
- true airspeed,  $V_{TAS}$  [kt];
- temperature deviation from standard atmosphere,  $\Delta T$  [K].

The equations corresponding to the different engine types are given below.

$$\text{Jet:} \quad (Thr_{\max \text{ climb}})_{ISA} = C_{Tc,1} \times \left( 1 - \frac{H_p}{C_{Tc,2}} + C_{Tc,3} \times H_p^2 \right) \quad (2.7-1)$$

$$\text{Turboprop:} \quad (Thr_{\max \text{ climb}})_{ISA} = \frac{C_{Tc,1}}{V_{TAS}} \times \left( 1 - \frac{H_p}{C_{Tc,2}} \right) + C_{Tc,3} \quad (2.7-2)$$

$$\text{Piston and Electric:} \quad (Thr_{\max \text{ climb}})_{ISA} = C_{Tc,1} \times \left( 1 - \frac{H_p}{C_{Tc,2}} \right) + \frac{C_{Tc,3}}{V_{TAS}} \quad (2.7-3)$$

For all engine types, the maximum climb thrust is corrected for temperature deviations from standard atmosphere,  $\Delta T$ , in the following manner:

$$Thr_{\max \text{ climb}} = (Thr_{\max \text{ climb}})_{ISA} \times (1 - C_{Tc,5} \cdot \Delta T_{\text{eff}}) \quad (2.7-4)$$

Where:

$$\Delta T_{\text{eff}} = \Delta T - C_{Tc,4} \quad (2.7-5)$$

with the limits:

$$0.0 \leq \Delta T_{\text{eff}} \times C_{Tc,5} \leq 0.4 \quad (2.7-6)$$

and:

$$C_{Tc,5} \geq 0.0 \quad (2.7-7)$$

This maximum climb thrust is used for both take-off and climb phases.

### 2.7.2. Maximum Cruise Thrust

The normal cruise thrust is by definition set equal to drag ( $Thr = D$ ). However, the maximum amount of thrust available in cruise situation is limited. The maximum cruise thrust is calculated as a ratio of the maximum climb thrust given by expression 2.7-4, that is:

$$(Thr_{cruise})_{MAX} = C_{Tcr} \times Thr_{max\ climb} \quad (2.7-8)$$

The coefficient  $C_{Tcr}$  is currently set uniformly for all aircraft (see Section 4.5).

### 2.7.3. Descent Thrust

Descent thrust is calculated as a ratio of the maximum climb thrust given by expression 2.7-4, with different correction factors used for high and low altitudes, and approach and landing configurations (see Section 2.5), that is:

$$\begin{aligned} \text{if } H_p > H_{p,des}: \\ Thr_{des,high} = C_{Tdes,high} \times Thr_{max\ climb} \end{aligned} \quad (2.7-9)$$

if  $H_p \leq H_{p,des}$ :

$$\text{Cruise configuration:} \quad Thr_{des,low} = C_{Tdes,low} \times Thr_{max\ climb} \quad (2.7-10)$$

$$\text{Approach configuration:} \quad Thr_{des,app} = C_{Tdes,app} \times Thr_{max\ climb} \quad (2.7-11)$$

$$\text{Landing configuration:} \quad Thr_{des,ld} = C_{Tdes,ld} \times Thr_{max\ climb} \quad (2.7-12)$$

Note that for those models where “non-clean” data (see Section 2.6.1) is available,  $H_{p,des}$  cannot be below  $H_{max,AP}$ .

## 2.8. REDUCED CLIMB POWER

The reduced climb power has been introduced to allow the simulation of climbs using less than the maximum climb setting. In day-to-day operations, many aircraft use a reduced setting during climb in order to extend engine life and save cost. The correction factors that are used to calculate the reduction in power have been obtained in an empirical way and have been validated with the help of air traffic controllers.

In BADA, climbs that are performed using the full climb power will result in profiles that match the reference data that is found in the Flight Manual of the aircraft. Climbs with reduced power will give a profile that may be more representative of actual operations.

$$C_{\text{pow,red}} = 1 - C_{\text{red}} \times \frac{m_{\text{max}} - m_{\text{act}}}{m_{\text{max}} - m_{\text{min}}} \quad (2.8-1)$$

The value of  $C_{\text{red}}$  is a function of the engine type and is given in Section 4.11.

Nevertheless:

If  $H_p < (0.8 \cdot h_{\text{max}})$ :

$C_{\text{red}} = f(\text{engine type})$  (see Section 4.11)

Else

$C_{\text{red}} = 0$  [dimensionless]

where  $h_{\text{max}}$  is given by expression 2.5-1.

The power reduction  $C_{\text{pow,red}}$  is to be applied during the climb phase in expression 2.2-7, which becomes:

$$\text{ROCD} = \frac{dH_p}{dt} = \frac{T - \Delta T}{T} \frac{(\text{Thr}_{\text{max climb}} - D) \cdot V_{\text{TAS}} \cdot C_{\text{pow,red}}}{m \cdot g_0} \cdot f\{M\} \quad (\text{in climb}) \quad (2.8-2)$$

## 2.9. FUEL CONSUMPTION

The BADA model provides coefficients that allow the calculation of the fuel flow for different engine types and regimes. The fuel flow is calculated in [kg/min] and includes the contribution from all engines. The subsections below provide the equations for each of the engine types and regimes.

### 2.9.1. Jet and Turboprop Engines

For the jet and turboprop engines, the thrust specific fuel consumption,  $\eta$  [kg/(min·kN)], is specified as a function of the true airspeed,  $V_{TAS}$  [kt]:

$$\text{jet:} \quad \eta = C_{f1} \times \left( 1 + \frac{V_{TAS}}{C_{f2}} \right) \quad (2.9-1)$$

$$\text{turboprop:} \quad \eta = C_{f1} \times \left( 1 - \frac{V_{TAS}}{C_{f2}} \right) \times \left( \frac{V_{TAS}}{1000} \right) \quad (2.9-2)$$

The nominal fuel flow,  $f_{nom}$  [kg/min], can then be calculated using the thrust, Thr:

$$\text{jet/turboprop:} \quad f_{nom} = \eta \times Thr \quad (2.9-3)$$

These expressions are used in all flight phases except during idle descent and cruise, where the following expressions are to be used.

The minimum fuel flow,  $f_{min}$  [kg/min], corresponding to idle thrust descent conditions for both jet and turboprop engines, is specified as a function of the geopotential pressure altitude,  $H_p$  [ft], that is:

$$\text{jet/turboprop:} \quad f_{min} = C_{f3} \left( 1 - \frac{H_p}{C_{f4}} \right) \quad (2.9-4)$$

Note that for both jet and turboprop engines, the idle thrust part of the descent stops when the aircraft switches to approach and landing configuration (see Section 2.5), at which point thrust is generally increased. Hence, the calculation of fuel flow during approach and landing phases shall be based on the nominal fuel flow (expressions 2.7-11, 2.7-12 and 2.9-3), and limited to the minimum fuel flow (expression 2.9-4) if necessary:

$$\text{jet/turboprop:} \quad f_{ap/ld} = \text{MAX} (f_{nom}, f_{min}) \quad (2.9-5)$$

The cruise fuel flow,  $f_{cr}$  [kg/min], is calculated using the thrust specific fuel consumption  $\eta$ , the thrust Thr, and a cruise fuel flow factor,  $C_{fcr}$ :

$$\text{jet/turboprop:} \quad f_{cr} = \eta \times Thr \times C_{fcr} \quad (2.9-6)$$

The cruise fuel flow correction factor has been established for a number of aircraft types whenever the reference data for cruise fuel consumption is available. This factor has been set to 1 (one) for all the other aircraft models.

### 2.9.2. Piston and Electric Engines

For piston and electric engines, the nominal fuel flow,  $f_{nom}$  [kg/min], is specified to be a constant, that is:

$$f_{nom} = C_{f1} \quad (2.9-7)$$

This expression is used in all flight phases except during descent and cruise, where the following expressions are to be used.

The minimum fuel flow,  $f_{min}$  [kg/min], corresponding to descent conditions for piston and electric engines, is specified to be a constant:

$$f_{min} = C_{f3} \quad (2.9-8)$$

The cruise fuel flow,  $f_{cr}$  [kg/min], is calculated using a cruise fuel flow factor,  $C_{fcr}$ :

$$f_{cr} = C_{f1} \times C_{fcr} \quad (2.9-9)$$

The cruise fuel flow correction factor has been established for a number of aircraft types whenever the reference data for cruise fuel consumption is available. This factor has been set to 1 (one) for all the other aircraft models.

Since electric engines do not use fuel, the values of the fuel coefficients for electric aircraft are determined so that the fuel flow is equal to zero.

## 2.10. GROUND MOVEMENT

Four values are specified that can be of use when simulating ground movements. These parameters are:

- TOL: FAR Take-Off Length [m] with MTOW on a dry, hard, level runway under ISA conditions and no wind.
- LDL: FAR Landing Length [m] with MLW on a dry, hard, level runway under ISA conditions and no wind.
- span: aircraft wingspan [m]
- length: aircraft length [m]

Note that currently the value of the MLW is not provided in BADA. Apart from these model-specific parameters, there are also a number of ground speeds defined as general parameters in Section 4.10.

## 2.11. SUMMARY OF OPERATIONS PERFORMANCE PARAMETERS

A summary of the parameters specified by the BADA operations performance model is supplied in Table 2-1 below. This table excludes those parameters that have been set to zero.

Detailed information on how these parameters have been obtained during the process of BADA aircraft model identification using the aircraft performance reference documents is provided in [RD4].

**Important notice:** Parameters listed in bold in the Table 2-1 below should not be modified by the user as such modifications may impact the validity of the model.

Table 2-1: BADA Operations Performance Parameter Summary

| Model Category                                                           | Symbols                                                                                                                                                                                                                                                                                  | Units                                                                                                                                                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aircraft type<br>(3 values)                                              | <b>n<sub>eng</sub></b><br><b>engine type</b><br><br><b>wake category</b>                                                                                                                                                                                                                 | dimensionless<br>string<br><br>string                                                                                                                                 | <b>number of engines</b><br><b>either Jet, Turboprop, Piston or Electric</b><br><br><b>either J, H, M or L</b>                                                                                                                                                                                                                                                                                                                                                        |
| Mass<br>(4 values)                                                       | <b>m<sub>ref</sub></b><br><b>m<sub>min</sub></b><br><b>m<sub>max</sub></b><br><b>m<sub>pyld</sub></b>                                                                                                                                                                                    | tonnes<br>tonnes<br>tonnes<br>tonnes                                                                                                                                  | <b>reference mass</b><br><b>minimum mass</b><br><b>maximum mass</b><br><b>maximum payload mass</b>                                                                                                                                                                                                                                                                                                                                                                    |
| Flight envelope<br>(6 values)                                            | <b>V<sub>MO</sub></b><br><b>M<sub>MO</sub></b><br><b>h<sub>MO</sub></b><br><b>h<sub>max</sub></b><br><b>G<sub>w</sub></b><br><b>G<sub>t</sub></b>                                                                                                                                        | knots (CAS)<br>dimensionless<br>feet<br>feet<br>feet/kg<br>feet/K                                                                                                     | <b>maximum operating speed</b><br><b>maximum operating Mach number</b><br><b>maximum operating altitude</b><br><b>max. altitude at MTOW and ISA</b><br><b>weight gradient on max. altitude</b><br><b>temperature gradient on max. altitude</b>                                                                                                                                                                                                                        |
| Aerodynamics<br>(15 values for Jet & TBP aircraft, 13 values for others) | <b>S</b><br><b>C<sub>D0,CR</sub></b><br><b>C<sub>D2,CR</sub></b><br><b>C<sub>D0,AP</sub></b><br><b>C<sub>D2,AP</sub></b><br><b>C<sub>D0,LD</sub></b><br><b>C<sub>D2,LD</sub></b><br><b>C<sub>D0,ALDG</sub></b><br><b>(V<sub>stall</sub>)<sub>i</sub></b><br><b>C<sub>Lbo</sub> (M=0)</b> | m <sup>2</sup><br>dimensionless<br>dimensionless<br>dimensionless<br>dimensionless<br>dimensionless<br>dimensionless<br>dimensionless<br>knots (CAS)<br>dimensionless | <b>reference wing surface area</b><br><b>parasitic drag coefficient (cruise)</b><br><b>induced drag coefficient (cruise)</b><br><b>parasitic drag coefficient (approach)</b><br><b>induced drag coefficient (approach)</b><br><b>parasitic drag coefficient (landing)</b><br><b>induced drag coefficient (landing)</b><br><b>parasite drag coef. (landing gear)</b><br><b>stall speed [TO, IC, CR, AP, LD]</b><br><b>buffet onset lift coef. (Jet &amp; TBP only)</b> |

| Model Category                | Symbols                                                                                                                                                                                                                                                                                                                                                             | Units                                                                                                                                                                                                                                                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | <b>k</b>                                                                                                                                                                                                                                                                                                                                                            | dimensionless                                                                                                                                                                                                                                                                           | <b>buffeting gradient (Jet &amp; TBP only)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Engine thrust<br>(12 values)  | <b>C<sub>Tc,1</sub></b><br><b>C<sub>Tc,2</sub></b><br><b>C<sub>Tc,3</sub></b><br><br><b>C<sub>Tc,4</sub></b><br><b>C<sub>Tc,5</sub></b><br><b>C<sub>Tdes,low</sub></b><br><b>C<sub>Tdes,high</sub></b><br><br><b>H<sub>p,des</sub></b><br><br><b>C<sub>Tdes,app</sub></b><br><b>C<sub>Tdes,ld</sub></b><br><b>V<sub>des,ref</sub></b><br><b>M<sub>des,ref</sub></b> | Newton (jet/piston/elec.)<br>knot-Newton (turboprop)<br>feet<br>1/feet <sup>2</sup> (jet)<br>Newton (turboprop)<br>knot-Newton<br>(piston/electric)<br><br>K<br>1/K<br>dimensionless<br>dimensionless<br><br>feet<br><br>dimensionless<br>dimensionless<br>knots (CAS)<br>dimensionless | <b>1st max. climb thrust coefficient</b><br><br><b>2nd max climb thrust coefficient</b><br><b>3rd max. climb thrust coefficient</b><br><br><br><b>1st thrust temperature coefficient</b><br><b>2nd thrust temperature coefficient</b><br><b>low altitude descent thrust coefficient</b><br><b>high altitude descent thrust coefficient</b><br><b>transition altitude for calculation of descent thrust</b><br><b>approach thrust coefficient</b><br><b>landing thrust coefficient</b><br>reference descent speed<br>reference descent Mach number |
| Fuel flow<br>(5 values)       | <b>C<sub>f1</sub></b><br><br><b>C<sub>f2</sub></b><br><br><b>C<sub>f3</sub></b><br><b>C<sub>f4</sub></b><br><b>C<sub>fcr</sub></b>                                                                                                                                                                                                                                  | kg/(min·kN) (jet)<br>kg/(min·kN·knot)<br>(turboprop)<br>kg/min (piston/electric)<br><br>knots<br><br>kg/min<br>feet<br>dimensionless                                                                                                                                                    | <b>1st thrust specific fuel consumption coefficient</b><br><br><br><b>2nd thrust specific fuel consumption coefficient</b><br><br><b>1st descent fuel flow coefficient</b><br><b>2nd descent fuel flow coefficient</b><br><b>cruise fuel flow correction coefficient</b>                                                                                                                                                                                                                                                                          |
| Ground movement<br>(4 values) | TOL<br>LDL<br>span<br>length                                                                                                                                                                                                                                                                                                                                        | m<br>m<br>m<br>m                                                                                                                                                                                                                                                                        | take-off length<br>landing length<br>wingspan<br>length                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

Note that the following coefficients can have negative values:

K, G<sub>t</sub>, C<sub>Tc,1</sub>, C<sub>Tc,2</sub>, C<sub>Tc,3</sub>, C<sub>Tc,4</sub>, C<sub>Tdes,low</sub>, C<sub>Tdes,high</sub>, C<sub>f1</sub>, C<sub>f2</sub>, C<sub>f4</sub>.

### 3. AIRLINE PROCEDURE MODELS

This section defines the standard airline procedures, which are parameterised by the BADA airline procedure models. Definition of the standard airline procedures in BADA is driven by a requirement to provide means of simulating standard or nominal aircraft operations using different simulation and modelling tools for various ATM applications.

The BADA airline procedure model is provided for three separate flight phases: climb, cruise and descent. For each of these phases and each aircraft model, the BADA airline procedure model requires the following information to determine the aircraft speed schedule:

1. BADA airline procedure default speeds provided in Airline Procedure File (APF):

V<sub>1</sub> - standard CAS [knots] below 10,000 ft;

V<sub>2</sub> - standard CAS [knots] between 10,000 ft and Mach transition altitude;

M - standard Mach number above Mach transition altitude;

where the Mach transition altitude is defined in Section 2.1.2 (g).

2. Stall speeds for take-off and landing configurations provided in Operations Performance File (OPF)
3. Coefficients provided in Sections 4.7 and 4.8.

The process of definition of the BADA airline procedure default speeds and choice of aircraft configurations in function of flight phase is described in [RD4]. The airline procedure model below 10,000 ft with corresponding coefficients (mentioned under item 3 above) have been defined taking into account aircraft manufacturers' performance reference data and aircraft operational data available at EUROCONTROL.

The fact that the way aircraft is operated varies significantly in function of specific airspace procedures and operating policies of locally dominant airlines is widely recognised. It is for that reason that the resulting speed schedules of the BADA standard airline procedure model may differ from a geographical location or of an aerospace's specific aircraft operation.

To account for the local aircraft operation characteristics and improve conformance of the simulated aircraft behaviour with real operations, the user of BADA is given a possibility to modify the BADA default speeds (as provided in APF file). The change of speed-related APF parameters should be done in accordance with the BADA modelling procedure described in the Chapter 2.2.3 of [RD4].

However, the stall speeds (as provided in OPF file) and coefficients detailed in Section 4.7 and 4.8 are not subject to modification. The BADA User should not modify them.

The altitude levels used for determination of CAS speed schedules and provided in the following chapters are expressed in terms of geopotential pressure altitude. However, different reference datums for altitude measurement<sup>9</sup> may be applied in function of the user application and its functional design choices.

The BADA Airline Procedure Model only identifies the possibility to introduce notion of different altitude altimetry for calculation of the CAS speed schedules in the user application. The implementation decision is left to the application owner.

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<sup>9</sup> Such as use of standard operational pressure settings used in aviation: QNH for MSL pressure, QFE for pressure at the airport reference point or QNE corresponding to standard MSL1013 hPa. These can be selected through the altimeter's pressure setting knob in the aircraft.

### 3.1. CLIMB

The following parameters are defined for each aircraft type to characterise the climb phase:

- $V_{cl,1}$  - standard climb CAS [knots] between 1,500/6,000 and 10,000 ft
- $V_{cl,2}$  - standard climb CAS [knots] between 10,000 ft and Mach transition altitude
- $M_{cl}$  - standard climb Mach number above Mach transition altitude

- For jet aircraft the following CAS schedule is assumed, based on the parameters mentioned above and the take-off stall speed:

$$\text{from 0 to 1,499 ft} \quad C_{Vmin} \cdot (V_{stall})_{TO} + V_{dCL,1} \quad (3.1-1)$$

$$\text{from 1,500 to 2,999 ft} \quad C_{Vmin} \cdot (V_{stall})_{TO} + V_{dCL,2} \quad (3.1-2)$$

$$\text{from 3,000 to 3,999 ft} \quad C_{Vmin} \cdot (V_{stall})_{TO} + V_{dCL,3} \quad (3.1-3)$$

$$\text{from 4,000 to 4,999 ft} \quad C_{Vmin} \cdot (V_{stall})_{TO} + V_{dCL,4} \quad (3.1-4)$$

$$\text{from 5,000 to 5,999 ft} \quad C_{Vmin} \cdot (V_{stall})_{TO} + V_{dCL,5} \quad (3.1-5)$$

$$\text{from 6,000 to 9,999 ft} \quad \min(V_{cl,1}, 250 \text{ kt})$$

$$\text{from 10,000 ft to Mach transition altitude} \quad V_{cl,2}$$

$$\text{above Mach transition altitude} \quad M_{cl}$$

- For turboprop, piston and electric aircraft the following CAS schedule is assumed:

$$\text{from 0 to 499 ft} \quad C_{Vmin} \cdot (V_{stall})_{TO} + V_{dCL,6} \quad (3.1-6)$$

$$\text{from 500 to 999 ft} \quad C_{Vmin} \cdot (V_{stall})_{TO} + V_{dCL,7} \quad (3.1-7)$$

$$\text{from 1,000 to 1,499 ft} \quad C_{Vmin} \cdot (V_{stall})_{TO} + V_{dCL,8} \quad (3.1-8)$$

$$\text{from 1,500 to 9,999 ft} \quad \min(V_{cl,1}, 250 \text{ kt})$$

$$\text{from 10,000 ft to Mach transition altitude} \quad V_{cl,2}$$

$$\text{above Mach transition altitude} \quad M_{cl}$$

Note 1: The take-off stall speed,  $(V_{stall})_{TO}$ , must be corrected for the difference in aircraft mass from the reference mass using formula 2.4-1. The values for  $V_{dCL,i}$  can be found in Section 4.8.

Note 2: The climb speed schedule shall determine an increasing speed from take-off to  $V_{cl,1}$ . To ensure that monotony, it is recommended to determine the speed schedule from the highest altitude to the lowest one, and to use at each step the speed of the higher altitude range as a ceiling value for the lower altitude range.

Note 3: Any speed from the schedule described above that would be lower (resp. higher) than the minimum (resp. maximum) speed determined for the same conditions using Section 2.5 (b) (resp. Section 2.5 (a)) shall be overridden by this minimum (resp. maximum) speed.

### 3.2. CRUISE

The following parameters are defined for each aircraft type to characterise the cruise phase:

- $V_{cr,1}$  - standard cruise CAS [knots] between 3,000 and 10,000 ft
- $V_{cr,2}$  - standard cruise CAS [knots] between 10,000 ft and Mach transition altitude
- $M_{cr}$  - standard cruise Mach number above Mach transition altitude

- For jet aircraft the following CAS schedule is assumed:

|                                            |                                  |
|--------------------------------------------|----------------------------------|
| from 0 to 2,999 ft                         | $\min(V_{cr,1}, 170 \text{ kt})$ |
| from 3,000 to 5,999 ft                     | $\min(V_{cr,1}, 220 \text{ kt})$ |
| from 6,000 to 13,999 ft                    | $\min(V_{cr,1}, 250 \text{ kt})$ |
| from 14,000 ft to Mach transition altitude | $V_{cr,2}$                       |
| above Mach transition altitude             | $M_{cr}$                         |

- For turboprop, piston and electric aircraft the following CAS schedule is assumed:

|                                            |                                  |
|--------------------------------------------|----------------------------------|
| from 0 to 2,999 ft                         | $\min(V_{cr,1}, 150 \text{ kt})$ |
| from 3,000 to 5,999 ft                     | $\min(V_{cr,1}, 180 \text{ kt})$ |
| from 6,000 to 9,999 ft                     | $\min(V_{cr,1}, 250 \text{ kt})$ |
| from 10,000 ft to Mach transition altitude | $V_{cr,2}$                       |
| above Mach transition altitude             | $M_{cr}$                         |

Note: Any speed from the schedule described above that would be lower (resp. higher) than the minimum (resp. maximum) speed determined for the same conditions using Section 2.5 (b) (resp. Section 2.5 (a)) shall be overridden by this minimum (resp. maximum) speed.

### 3.3. DESCENT

The following parameters are defined for each aircraft type to characterise the descent phase:

- $V_{des,1}$  - standard descent CAS [knots] between 3,000/6,000 and 10,000 ft
- $V_{des,2}$  - standard descent CAS [knots] between 10,000 ft and Mach transition altitude
- $M_{des}$  - standard descent Mach number above Mach transition altitude

- For jet and turboprop aircraft the following CAS schedule is assumed, based on the above parameters and the landing stall speed:

$$\text{from 0 to 999 ft} \quad C_{Vmin} \cdot (V_{stall})_{LD} + V_{dDES,1} \quad (3.3-1)$$

$$\text{from 1,000 to 1,499 ft} \quad C_{Vmin} \cdot (V_{stall})_{LD} + V_{dDES,2} \quad (3.3-2)$$

$$\text{from 1,500 to 1,999 ft} \quad C_{Vmin} \cdot (V_{stall})_{LD} + V_{dDES,3} \quad (3.3-3)$$

$$\text{from 2,000 to 2,999 ft} \quad C_{Vmin} \cdot (V_{stall})_{LD} + V_{dDES,4} \quad (3.3-4)$$

$$\text{from 3,000 to 5,999 ft} \quad \min(V_{des,1}, 220)$$

$$\text{from 6,000 to 9,999 ft} \quad \min(V_{des,1}, 250)$$

$$\text{from 10,000 ft to Mach transition altitude} \quad V_{des,2}$$

$$\text{above Mach transition altitude} \quad M_{des}$$

- For piston and electric aircraft the following CAS schedule is assumed:

$$\text{from 0 to 499 ft} \quad C_{Vmin} \cdot (V_{stall})_{LD} + V_{dDES,5} \quad (3.3-5)$$

$$\text{from 500 to 999 ft} \quad C_{Vmin} \cdot (V_{stall})_{LD} + V_{dDES,6} \quad (3.3-6)$$

$$\text{from 1000 to 1,499 ft} \quad C_{Vmin} \cdot (V_{stall})_{LD} + V_{dDES,7} \quad (3.3-7)$$

$$\text{from 1,500 to 9,999 ft} \quad V_{des,1}$$

$$\text{from 10,000 ft to Mach transition altitude} \quad V_{des,2}$$

$$\text{above Mach transition altitude} \quad M_{des}$$

Note 1: The landing stall speed,  $(V_{stall})_{LD}$ , must be corrected for the difference in aircraft mass from the reference mass using formula 2.4-1. The values for  $V_{dDES,i}$  can be found in Section 4.8.

Note 2: The descent speed schedule shall determine a decreasing speed from  $V_{des,1}$  to landing. To ensure that monotony, it is recommended to evaluate the speed schedule from the highest altitude to the lowest one, and to use at each step the speed of the higher altitude range as a ceiling value for the lower altitude range.

Note 3: Any speed from the schedule described above that would be lower (resp. higher) than the minimum (resp. maximum) speed determined for the same conditions using Section 2.5 (b) (resp. Section 2.5 (a)) shall be overridden by this minimum (resp. maximum) speed.

## 4. GLOBAL AIRCRAFT PARAMETERS

### 4.1. INTRODUCTION

A number of parameters that have been described in Section 2 (Operations Performance Model) and Section 3 (Airline Procedure Model) have values that are independent of the aircraft type or model for which they are used. The values of these and other parameters which have general use have been put in the Global Parameters File (BADA.GPF). This increases the flexibility and allows an easier evaluation of the values that are used.

The next section gives an overview of the parameters that are defined in the Global Parameters File. If relevant, it also indicates the formula in which the parameter should be used.

### 4.2. MAXIMUM ACCELERATION

Maximum acceleration parameters are used to limit the increment in TAS (longitudinal) or ROCD (normal). Two parameters are defined:

| Name:             | Description:                                        | Value [ft/s <sup>2</sup> ]: |
|-------------------|-----------------------------------------------------|-----------------------------|
| $a_{l,max} (civ)$ | maximum longitudinal acceleration for civil flights | 2.0                         |
| $a_{n,max} (civ)$ | maximum normal acceleration for civil flights       | 5.0                         |

The two acceleration limits are to be used in the following way:

$$\text{longitudinal acceleration: } |V_k - V_{k-1}| \leq a_{l,max} (civ) \Delta t \quad (4.2-1)$$

$$\text{normal acceleration: } |\gamma_k - \gamma_{k-1}| \leq \frac{a_{n,max} (civ) \Delta t}{V} \quad (4.2-2)$$

where,

$$\gamma = \sin^{-1} \left( \frac{\dot{h}}{V} \right) \quad (4.2-3)$$

and,

|            |                                                     |
|------------|-----------------------------------------------------|
| $\gamma$   | is the climb/descent angle,                         |
| $V$        | is the true airspeed [ft/s],                        |
| $k, k-1$   | indicate values at update intervals $k$ and $k-1$ , |
| $\Delta t$ | is the time interval between $k$ and $k-1$ [s]      |

The values for the maximum longitudinal acceleration for military flights,  $a_{l,max} (mil)$ , and for the maximum normal acceleration for military flights,  $a_{n,max} (mil)$ , are currently undefined.

### 4.3. BANK ANGLES

Nominal and maximum bank angles are defined separately for military and civil flights. These bank angles can be used to calculate nominal and maximum rate of turns.

| Name:                            | Description:                                                 | Value [deg]: |
|----------------------------------|--------------------------------------------------------------|--------------|
| $\phi_{\text{nom,civ (TO,LD)}}$  | Nominal bank angles for civil flight during TO and LD        | 15           |
| $\phi_{\text{nom,civ (OTHERS)}}$ | Nominal bank angles for civil flight during all other phases | 30           |
| $\phi_{\text{nom,mil}}$          | Nominal bank angles for military flight (all phases)         | 50           |
| $\phi_{\text{max,civ (TO,LD)}}$  | Maximum bank angles for civil flight during TO and LD        | 25           |
| $\phi_{\text{max,civ (HOLD)}}$   | Maximum bank angles for civil flight during HOLD             | 35           |
| $\phi_{\text{max,civ (OTHERS)}}$ | Maximum bank angles for civil flight during all other phases | 45           |
| $\phi_{\text{max,mil}}$          | Maximum bank angles for military flight (all phases)         | 70           |

The rate of turn,  $\dot{\phi}$ , is calculated as a function of the bank angle:

$$\dot{\phi} = \frac{g_0}{V_{\text{TAS}}} \times \tan(\phi) \quad (4.3-1)$$

### 4.4. EXPEDITED DESCENT

The expedited descent factor is to be used as a drag multiplication factor during expedited descents in order to simulate the use of spoilers:

| Name:                | Description:             | Value [ - ]: |
|----------------------|--------------------------|--------------|
| $C_{\text{des,exp}}$ | Expedited descent factor | 1.6          |

The drag during an expedited descent is calculated using the nominal drag (see Section 2.6.1):

$$D_{\text{des,exp}} = C_{\text{des,exp}} \cdot D_{\text{nom}} \quad (4.4-1)$$

### 4.5. THRUST FACTORS

Maximum take-off and maximum cruise thrust factors have been specified. The  $C_{\text{Th,TO}}$  factor is no longer in use (since BADA 3.0). The  $C_{\text{Tcr}}$  factor is to be used in expression 2.7-8.

| Name:              | Description:                      | Value [ - ]: |
|--------------------|-----------------------------------|--------------|
| $C_{\text{Th,TO}}$ | Take-off thrust coefficient       | 1.2          |
| $C_{\text{Tcr}}$   | Maximum cruise thrust coefficient | 0.95         |

#### 4.6. CONFIGURATION ALTITUDE THRESHOLD

Altitude thresholds have been specified in BADA for four configurations: take-off (TO), initial climb (IC), approach (AP) and landing (LD). Note that the selection of the take-off and initial climb configurations is defined only with the altitude. The selection of the approach and landing configurations is done through the use of air speed and altitude (see Section 2.5), while the altitudes at which the configuration change takes place should not be higher than the ones given below. The altitude values are expressed in terms of geopotential pressure altitude.

| <b>Name:</b>  | <b>Description:</b>                          | <b>Value [ft]:</b> |
|---------------|----------------------------------------------|--------------------|
| $H_{\max,TO}$ | Maximum altitude threshold for take-off      | 400                |
| $H_{\max,IC}$ | Maximum altitude threshold for initial climb | 2,000              |
| $H_{\max,AP}$ | Maximum altitude threshold for approach      | 8,000              |
| $H_{\max,LD}$ | Maximum altitude threshold for landing       | 3,000              |

#### 4.7. MINIMUM SPEED COEFFICIENTS

Two minimum speed coefficients are specified, which are to be used in expressions 2.5-2 and 2.5-3 and (for  $C_{Vmin}$  only) in Sections 3.1 and 3.3:

| <b>Name:</b>  | <b>Description:</b>                          | <b>Value [ - ]:</b> |
|---------------|----------------------------------------------|---------------------|
| $C_{Vmin,TO}$ | Minimum speed coefficient for take-off       | 1.2                 |
| $C_{Vmin}$    | Minimum speed coefficient (all other phases) | 1.3                 |

#### 4.8. SPEED SCHEDULES

The speed schedules applicable below FL100 for climb and descent are based on a factored stall speed plus increment valid for a specified geopotential pressure altitude range.

| <b>Name:</b> | <b>Description:</b>                                         | <b>Value [KCAS]:</b> |
|--------------|-------------------------------------------------------------|----------------------|
| $V_{dCL,1}$  | Climb speed increment below 1500 ft (jet)                   | 5                    |
| $V_{dCL,2}$  | Climb speed increment below 3000 ft (jet)                   | 10                   |
| $V_{dCL,3}$  | Climb speed increment below 4000 ft (jet)                   | 30                   |
| $V_{dCL,4}$  | Climb speed increment below 5000 ft (jet)                   | 60                   |
| $V_{dCL,5}$  | Climb speed increment below 6000 ft (jet)                   | 80                   |
| $V_{dCL,6}$  | Climb speed increment below 500 ft (turbo/piston/electric)  | 20                   |
| $V_{dCL,7}$  | Climb speed increment below 1000 ft (turbo/piston/electric) | 30                   |
| $V_{dCL,8}$  | Climb speed increment below 1500 ft (turbo/piston/electric) | 35                   |
| $V_{dDES,1}$ | Descent speed increment below 1000 ft (jet/turboprop)       | 5                    |
| $V_{dDES,2}$ | Descent speed increment below 1500 ft (jet/turboprop)       | 10                   |
| $V_{dDES,3}$ | Descent speed increment below 2000 ft (jet/turboprop)       | 20                   |
| $V_{dDES,4}$ | Descent speed increment below 3000 ft (jet/turboprop)       | 50                   |
| $V_{dDES,5}$ | Descent speed increment below 500 ft (piston/electric)      | 5                    |

|             |                                                         |    |
|-------------|---------------------------------------------------------|----|
| $V_{DES,6}$ | Descent speed increment below 1000 ft (piston/electric) | 10 |
| $V_{DES,7}$ | Descent speed increment below 1500 ft (piston/electric) | 20 |

These values are to be used in the expressions in Sections 3.1 and 3.3.

#### 4.9. HOLDING SPEEDS

The holding speeds that are to be used to calculate holding areas are defined according to the ICAO standards:

| Name:        | Description:                          | Value [KCAS]: |
|--------------|---------------------------------------|---------------|
| $V_{hold,1}$ | Holding speed below FL140             | 230           |
| $V_{hold,2}$ | Holding speed between FL140 and FL200 | 240           |
| $V_{hold,3}$ | Holding speed between FL200 and FL340 | 265           |
| $V_{hold,4}$ | Holding speed above FL340 [Mach]      | 0.83          |

Note that the holding speeds that are used by individual aircraft may vary between types.

#### 4.10. GROUND SPEEDS

A number of ground speeds are defined for the simulation of ground movement. For the moment, no distinction between aircraft type or engine type is made. The following speeds have been defined:

| Name:           | Description:           | Value [KCAS]: |
|-----------------|------------------------|---------------|
| $V_{backtrack}$ | Runway backtrack speed | 35            |
| $V_{taxi}$      | Taxi speed             | 15            |
| $V_{apron}$     | Apron speed            | 10            |
| $V_{gate}$      | Gate speed             | 5             |

The runway backtrack speed is the speed the aircraft will maintain when it backtracks across the runway. The taxi speed is used anywhere between the runway and the apron area. The apron speed is used in the apron area while the gate speed is used for all manoeuvring between the gate position and the apron.

#### 4.11. REDUCED POWER COEFFICIENT

The reduced power coefficients are defined for the different engine types. It is stressed that the values given below were found in an empirical way and have been validated with the help of air traffic controllers:

| Name:            | Description:                             | Value [ - ]: |
|------------------|------------------------------------------|--------------|
| $C_{red,jet}$    | Maximum reduction in power for jet       | 0.15         |
| $C_{red,turbo}$  | Maximum reduction in power for turboprop | 0.25         |
| $C_{red,piston}$ | Maximum reduction in power for piston    | 0.0          |
| $C_{red,elec}$   | Maximum reduction in power for electric  | 0.0          |

The coefficients should be used in Formula 2.8-1.

## 5. FILE STRUCTURE

BADA Family 3 provides operations and procedures data for various aircraft types. For some of these aircraft types, the data are provided directly in specific files. These aircraft types are considered as directly supported, and referred to as aircraft original models. The way they have been identified is described in [RD4]. For the other aircraft types, the data are specified to be the same as one of the directly supported aircraft types. These aircraft types are considered as 'equivalent' to one of the original aircraft models, and referred to as synonym aircraft.

### 5.1. FILE TYPES

All data provided by BADA Family 3 are organised into six types of files:

- one Synonym File,
  - a set of Operations Performance Files,
  - a set of Airline Procedure Files,
  - a set of Performance Table Files,
  - a set of Performance Table Data,
  - a Global Parameter File.
- The Synonym File, named SYNONYM.NEW, provides a list of all the aircraft types which are supported by BADA and indicates whether each aircraft type is supported directly (through provision of parameters in other files) or supported by equivalence (through indicating an equivalent aircraft type that is supported directly). In addition, the file indicates whether the aircraft type is recognised by ICAO in [RD1]. The format of the file is described in Section 5.3.
  - There is one Operations Performance File (OPF) provided for each directly supported aircraft type. This file specifies parameter values for the mass, flight envelope, drag, engine thrust and fuel consumption that are described in Section 2. Details on the format of the OPF file are given in Section 5.4.
  - There is one Airline Procedures File (APF) for each directly supported aircraft type. This file specifies the nominal manoeuvre speeds that are described in Section 3. Details on the format of the APF file are given in Section 5.5.
  - There is one Performance Table File (PTF) for each directly supported aircraft type. This file contains a summary table of speeds, climb/descent rates and fuel consumption at various flight levels. Details on the format of the PTF file are given in Section 5.6.
  - There is one Performance Table Data (PTD) file for each directly supported aircraft type. This file contains a detailed table of computed performance values at various flight levels. Details on the format of the PTD file are given in Section 5.7.
  - Finally there is one Global Parameter File which is named BADA.GPF. This file contains parameters that are described in Section 4 and are valid for all aircraft or a group of aircraft (for instance all civil flights or all jet aircraft). Details on the format of the GPF file are given in Section 5.8.

The names of the OPF, APF, PTF and PTD files are based on the ICAO designation code for the aircraft type. Except for some generic military fighter aircraft types (FGTH, FGTL), this code is the same as the International Civil Aviation Organisation (ICAO) designator code for the aircraft type [RD1]. That is:

|                                   |                   |
|-----------------------------------|-------------------|
| Operations Performance File name: | <ICAO_code>__.OPF |
| Airline Procedures File name:     | <ICAO_code>__.APF |
| Performance Table File name:      | <ICAO_code>__.PTF |
| Performance Table Data name:      | <ICAO_code>__.PTD |

Note that there are at least two underscore characters between the ICAO code and the file extension such that the length of the file name without the extension is six characters. Most ICAO codes are four characters in length and thus have two underscore characters. Some ICAO codes, however, can be shorter and thus require more underscore characters.

## 5.2. RELEASE SUMMARY FILE

The ReleaseSummary file provides a list of all files delivered as part of the BADA release. It lists, for each BADA file, the file name and BADA release identification, which is the BADA release in which the file was last modified.

### 5.3. SYNONYM FILE FORMAT

The SYNONYM.NEW file is an ASCII file, which lists all aircraft types supported by the BADA revision. All supported aircraft are listed alphabetically in the file whether they are supported directly or by equivalence. An example of the SYNONYM.NEW file is given below (partial listing).

```

CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC SYNONYM.NEW CCCCCCCCCCCCCCCCC/
CC                                                                                               /
CC               BADA SYNONYM FILE                                                                /
CC                                                                                               /
CC               File_name: SYNONYM.NEW                                                            /
CC                                                                                               /
CC               Creation_date: Apr 30 2002                                                         /
CC                                                                                               /
CC               Modification_date: Nov 24 2020                                                     /
CC                                                                                               /
CC===== Aircraft List =====                                                                    /
CC                                                                                               /
CC   A/C      MANUFACTURER      NAME OR MODEL      FILE      ICAO /
CC   CODE                                           /
CC                                                                                               /
CD * A306     AIRBUS             A300B4-622         J2H_____ Y /
CD * A30B     AIRBUS             A300B4-203         J2H_____ Y /
CD * A310     AIRBUS             A310-204         J2H_____ Y /
CD * A318     AIRBUS             A318-112         J2M_____ Y /
CD * A319     AIRBUS             A319-131         J2M_____ Y /
CD * A320     AIRBUS             A320-231         J2M_____ Y /
CD * A321     AIRBUS             A321-111         J2M_____ Y /
CD * A332     AIRBUS             A330-243         J2H_____ Y /
CD * A333     AIRBUS             A330-301         J2H_____ Y /
CD * A342     AIRBUS             A340-213         J4H_____ Y /
CD * A343     AIRBUS             A340-313         J4H_____ Y /
CD * A345     AIRBUS             A340-541         J4H_____ Y /
CD * A346     AIRBUS             A340-642         J4H_____ Y /
CD * A388     AIRBUS             A380-841         J4H_____ Y /
CD * AT43     ATR                42-300          TP2M_____ Y /
CD * AT44     ATR                42-400          TP2M_____ Y /
CD * AT45     ATR                42-500          TP2M_____ Y /
CD * AT46     ATR                42-600          TP2M_____ Y /
CD * AT72     ATR                72-200          TP2M_____ Y /
CD * AT73     ATR                72-210          TP2M_____ Y /
CD * AT75     ATR                72-210A         TP2M_____ Y /
CD * AT76     ATR                72-212A 600     TP2M_____ Y /
CD * B733     BOEING             B737-300         J2M_____ Y /
CD * B734     BOEING             B737-400e        J2M_____ Y /
CD * B735     BOEING             B737-500         J2M_____ Y /
CD * B736     BOEING             B737-600         J2M_____ Y /

```

There are three types of lines in the SYNONYM.NEW file with the line type identified by the first two characters in the line. These line types with their associated two leading characters are listed below.

|    |                  |
|----|------------------|
| CC | comment line     |
| CD | data line        |
| FI | end-of-file line |

The data is organised into two blocks separated by a comment line consisting of the block name and equal signs "=":

- file identification block
- aircraft list block

Each of these blocks is described in the subsections below.

### 5.3.1.1. File Identification Block

The file identification block provides information on the file name, creation and modification date. The block consists of 12 comment lines as shown below.

```

CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC SYNONYM.NEW CCCCCCCCCCCCCCCC/
CC                                          /
CC          BADA SYNONYM FILE                      /
CC                                          /
CC          File_name: SYNONYM.NEW                  /
CC                                          /
CC          Creation_date: Apr 30 2002              /
CC                                          /
CC          Modification_date: Nov 24 2020          /
CC                                          /
CC                                          /
CC                                          /

```

The comment lines specify the file name along with the creation and last modification date. The creation date indicates the date when the file was created for the first time. The modification date indicates when the contents of the file were last modified.

### 5.3.1.2. Aircraft List Block

The aircraft list block consists of 5 comment lines and one data line for each aircraft supported by the BADA revision. A partial listing of this block is shown below.

```

CC===== Aircraft List =====/
CC                                          /
CC  A/C      MANUFACTURER      NAME OR MODEL      FILE      ICAO /
CC  CODE                                          /
CC                                          /
CD * A320    AIRBUS            A320-231      J2M_____ Y /
CD * A346    AIRBUS            A340-642      J4H_____ Y /
CD * AT45    ATR                42-500        TP2M_____ Y /
CD * B773    BOEING            B777-300      J2H_____ Y /
CD - BZJT    (GENERIC)         Business jet   BZJT_____ N /
CD * C172    CESSNA            172S Skyhawk SP  GA_____ Y /

```

Each data line consists of 6 fields as described below:

#### (a) Support Type Field

This field is one character in length being one of the following two values:

- "-" to indicate an aircraft type directly supported, and,
- "" to indicate an aircraft type supported by equivalence with another directly supported aircraft

#### (b) Aircraft Code Field

This field identifies the aircraft type designator. It consists of a two- to four-character ICAO code.

#### (c) Manufacturer Field

This field identifies the manufacturer of the aircraft.

#### (d) Name or Model Field

This field identifies the name or model for the aircraft type.

(e) File Field

This field indicates the name of the OPF, APF, PTF and PTD files, which contain the parameters for the aircraft type (minus the file extension).

For an aircraft type which is directly supported, this file name will be the same as the ICAO code with an additional two or more underscore characters to form a string of six characters in length.

For an aircraft type which is supported through equivalence, the file name will indicate the file for the equivalent aircraft type which should be used.

(f) ICAO Field

This field indicates whether the designator for this aircraft type is in use (value “Y”) or not (value “N”) according to ICAO Doc 8643 [RD1]. This allows the BADA user to either conform to the latest ICAO Doc 8643, or maintain compatibility with legacy data. Aircraft types associated with the “N” value include generic BADA models whose names are not official ICAO designators, such as FGTL or HELI, and decommissioned aircraft types, such as MIR4 or NIM.

The above fields are specified in the following fixed format (Fortran notation):

'CD', 1X, A1, 1X, A4, 3X, A18, 1X, A25, 1X, A6, 2X, A1

## 5.4. OPF FILE FORMAT

The Operations Performance File (OPF) is an ASCII file, which for a particular aircraft type specifies the operations performance parameters described in Section 2. An example of an OPF file for a dummy jet aircraft is shown below.

```

CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC J2H____.OPF CCCCCCCCCCCCCCCCCCCCCCCCCCCCCC/
CC                                                                                   /
CC               AIRCRAFT PERFORMANCE OPERATIONAL FILE                           /
CC                                                                                   /
CC       File_name: J2H____.OPF                                                  /
CC                                                                                   /
CC       Creation_date: Apr 30 2002                                              /
CC                                                                                   /
CC       Modification_date: Sep 05 2008                                          /
CC                                                                                   /
CC===== Actype =====/
CD   J2H____          2 engines      Jet                                     H      /
CC   Heavy twin jet with XXX engines                                           wake /
CC                                                                                   /
CC===== Mass (t) =====/
CC   reference      minimum      maximum      max payload  mass grad /
CD   .14000E+03     .87000E+02     .17170E+03     .39000E+02     .15103E+00 /
CC===== Flight envelope =====/
CC   VMO(KCAS)      MMO          Max.Alt      Hmax          temp grad /
CD   .33500E+03     .82000E+00     .41000E+05     .32378E+05     -.2716E+02 /
CC===== Aerodynamics =====/
CC Wing Area and Buffet coefficients (SIM) /
CCndrst Surf(m2)    Clbo(M=0)      k          CM16 /
CD 5 .26000E+03     .13150E+01     .84080E+00     .00000E+00 /
CC Configuration characteristics /
CC n Phase Name     Vstall(KCAS)    CD0          CD2          unused /
CD 1 CR   Clean      .15100E+03     .20591E-01     .51977E-01     .00000E+00 /
CD 2 IC   S15F00     .11700E+03     .33057E-01     .45362E-01     .00000E+00 /
CD 3 TO   S15F00     .11700E+03     .33057E-01     .45362E-01     .00000E+00 /
CD 4 AP   S15F15     .10900E+03     .38031E-01     .44932E-01     .00000E+00 /
CD 5 LD   S30F40     .97000E+02     .78935E-01     .44822E-01     .00000E+00 /
CC Spoiler /
CD 1 RET /
CD 2 EXT .00000E+00 .00000E+00 /
CC Gear /
CD 1 UP /
CD 2 DOWN .22500E-01 .00000E+00 .00000E+00 /
CC Brakes /
CD 1 OFF /
CD 2 ON .00000E+00 .00000E+00 /
CC===== Engine Thrust =====/
CC Max climb thrust coefficients (SIM) /
CD .29716E+06 .51306E+05 .56296E-10 .84814E+01 .44597E-02 /
CC Desc(low) Desc(high) Desc level Desc(app) Desc(ld) /
CD .32012E-01 .40310E-01 .15161E+05 .13124E+00 .39136E+00 /
CC Desc CAS Desc Mach unused unused unused /
CD .30000E+03 .78000E+00 .00000E+00 .00000E+00 .00000E+00 /
CC===== Fuel Consumption =====/
CC Thrust Specific Fuel Consumption Coefficients /
CD .63936E+00 .10047E+04 /
CC Descent Fuel Flow Coefficients /
CD .21196E+02 .67071E+05 /
CC Cruise Corr. unused unused unused unused /
CD .98852E+00 .00000E+00 .00000E+00 .00000E+00 .00000E+00 /
CC===== Ground =====/
CC TOL LDL span length unused /
CD .23620E+04 .15550E+04 .44840E+02 .54080E+02 .00000E+00 /
CC===== /
FI /

```

There are three types of lines in the OPF file with the line type identified by the first two characters in the line. These line types with their associated two leading characters are listed below.

|    |                  |
|----|------------------|
| CC | comment line     |
| CD | data line        |
| FI | end-of-file line |

The comment lines are provided solely for the purpose of improving the readability of the file. All coefficients are contained within the CD lines in a fixed format. The end-of-file line is included as the last line in the file in order to facilitate the reading of the file in certain computing environments.

The data is organised into a total of eight blocks with each block separated by a comment line containing the block name and equal signs "=". These blocks are listed below and are described in further detail in the subsections below.

- file identification block
- aircraft type block
- mass block
- flight envelope block
- aerodynamics block
- engine thrust block,
- fuel consumption block
- ground movements block

### 5.4.1. File Identification Block

The file identification block provides information on the file name, creation date and modification date. The block consists of 12 comment lines. An example of the file identification block is shown below.

```

CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC J2H____.OPF CCCCCCCCCCCCCCCCCCCCCCCCCCCCCC/
CC                                                                                   /
CC               AIRCRAFT PERFORMANCE OPERATIONAL FILE                           /
CC                                                                                   /
CC                                                                                   /
CC      File_name: J2H____.OPF                                                    /
CC                                                                                   /
CC      Creation_date: Apr 30 2002                                                /
CC                                                                                   /
CC      Modification_date: Sep 05 2008                                           /
CC                                                                                   /
CC                                                                                   /

```

The comment lines specify the file name along with the creation date and last modification date. The creation date indicates the date when the file was created for the first time. The modification date indicates when the contents of the file were last modified.

### 5.4.2. Aircraft Type Block

The OPF aircraft type block consists of 1 data line with 3 comment lines for a total of 4 lines. An example of the aircraft type block is given below.

```

CC===== Actype =====/
1 -> CD   J2H____      2 engines      Jet      H      /
CC   Heavy twin jet with XXX engines      wake      /
CC                                                                                   /

```

The data line specifies the following aircraft type parameters:

- ICAO aircraft code (followed by 2 or more underscore characters as required to form a six-character string)
- number of engines,  $n_{eng}$
- engine type
- wake category

The engine type can be one of the following four values: Jet, Turboprop, Piston or Electric. The wake category can be one of the four values J (jumbo), H (heavy), M (medium) or L (light).

The four values are specified in the following fixed format (Fortran notation)

'CD', 3X, A6, 9X, I1, 12X, A9, 17X, A1

The comment lines typically indicate the aircraft model number and name and the engine designation.

### 5.4.3. Mass Block

The OPF mass block consists of 1 data line with 2 comment lines for a total of 3 lines.

An example of the mass block is given below.

```

CC===== Mass (t) =====/
CC      reference      minimum      maximum      max payload  mass grad  /
1 ->  CD      .14000E+03  .87000E+02  .17170E+03  .39000E+02  .15103E+00  /

```

The data line specifies the following BADA mass model parameters:

$m_{ref}$                    $m_{min}$                    $m_{max}$                    $m_{pyld}$                    $G_w$

These parameters are specified in the following fixed format (Fortran notation)

'CD', 2X, 5 (3X, E10.5)

### 5.4.4. Flight Envelope Block

The OPF flight envelope block consists of 1 data line with 2 comment lines for a total of 3 lines. An example of the flight envelope block is given below.

```

CC===== Flight envelope =====/
CC      VMO(KCAS)      MMO      Max.Alt      Hmax      temp grad  /
1 ->  CD      .33500E+03  .82000E+00  .41000E+05  .32378E+05  -.2716E+02  /

```

The data line specifies the following BADA speed envelope parameters:

$V_{MO}$                    $M_{MO}$                    $h_{MO}$                    $h_{max}$                    $G_t$

Note that all altitudes are expressed in feet.

These parameters are specified in the following fixed format (Fortran notation):

'CD', 2X, 5 (3X, E10.5)

### 5.4.5. Aerodynamics Block

The OPF aerodynamics block consists of 12 data lines and 8 comment lines for a total of 20 lines. An example of the aerodynamics block is given below.

```

CC===== Aerodynamics =====/
CC Wing Area and Buffet coefficients (SIM) /
CCndrst Surf(m2) Clbo(M=0) k CM16 /
1 -> CD 5 .26000E+03 .13150E+01 .84080E+00 .00000E+00 /
CC Configuration characteristics /
CC n Phase Name Vstall(KCAS) CD0 CD2 unused /
2 -> CD 1 CR Clean .15100E+03 .20591E-01 .51977E-01 .00000E+00 /
3 -> CD 2 IC S15F00 .11700E+03 .33057E-01 .45362E-01 .00000E+00 /
4 -> CD 3 TO S15F00 .11700E+03 .33057E-01 .45362E-01 .00000E+00 /
5 -> CD 4 AP S15F15 .10900E+03 .38031E-01 .44932E-01 .00000E+00 /
6 -> CD 5 LD S30F40 .97000E+02 .78935E-01 .44822E-01 .00000E+00 /
CC Spoiler /
7 -> CD 1 RET /
8 -> CD 2 EXT .00000E+00 .00000E+00 /
CC Gear /
9 -> CD 1 UP /
10 -> CD 2 DOWN .22500E-01 .00000E+00 .00000E+00 /
CC Brakes /
11 -> CD 1 OFF /
12 -> CD 2 ON .00000E+00 .00000E+00 /

```

The first data line specifies the following BADA aerodynamic model parameters:

S                       $C_{lbo}(M=0)$                       k                       $C_{M16}$

These parameters are specified in the following fixed format (Fortran notation):

'CD', 2X, 4 (3X, E10.5)

Note that the ndrst (number of drag settings) and  $C_{M16}$  parameters are not used but are included for backward compatibility reasons.

The next line holds the stall speed and flap setting for cruise as well as the values for the two drag coefficients for this configuration:

$(V_{stall})_{CR}$                        $C_{D0}$                        $C_{D2}$

These parameters are specified in the following fixed format (Fortran notation):

'CD', 15X, 3 (3X, E10.5)

The next four data lines have the same format and correspond to the other configurations. The configurations are specified in the following order, corresponding to a semi-monotonically decreasing stall speed:

IC      initial climb  
TO      take-off  
AP      approach  
LD      landing

The stall speed,  $(V_{stall})_i$ , is specified for each configuration, and  $C_{D0}$  and  $C_{D2}$  are given if available in the following fixed format (Fortran notation):

'CD', 15X, 3 (3X, E10.5)

In case the IC configuration is equal to the CR configuration, the values for  $C_{D0}$  and  $C_{D2}$  are mentioned only in the CR dataline. Note that the  $C_{D0}$  and  $C_{D2}$  coefficients for IC and TO configurations are not used but are included for backward compatibility reasons.

The data lines 7 through 9 are not used but are included for backward compatibility reasons.

Dataline 10 holds the drag increment for landing gear down:

$$C_{D0,ALDG}$$

The format of this line is:

'CD', 31X, E10.5

Datalines 11 and 12 are not used but are included for backward compatibility reasons.

#### 5.4.6. Engine Thrust Block

The OPF engine thrust block consists of 3 data lines with 4 comment lines for a total of 7 lines. An example of the engine thrust block is given below.

```
CC===== Engine Thrust =====/
CC      Max climb thrust coefficients (SIM) /
1 ->  CD      .29716E+06  .51306E+05  .56296E-10  .84814E+01  .44597E-02 /
CC      Desc(low)  Desc(high)  Desc level  Desc(app)  Desc(ld) /
2 ->  CD      .32012E-01  .40310E-01  .15161E+05  .13124E+00  .39136E+00 /
CC      Desc CAS   Desc Mach   unused      unused      unused /
3 ->  CD      .30000E+03  .78000E+00  .00000E+00  .00000E+00  .00000E+00 /
```

The first data line specifies the following BADA parameters used to calculate the maximum climb thrust:

$$C_{Tc,1} \quad C_{Tc,2} \quad C_{Tc,3} \quad C_{Tc,4} \quad C_{Tc,5}$$

These parameters are specified in the following fixed format (Fortran notation):

'CD', 2X, 5 (3X, E10.5)

The second data line specifies the following BADA parameters used to calculate cruise and descent thrust:

$$C_{Tdes,low} \quad C_{Tdes,high} \quad H_{p,des} \quad C_{Tdes,app} \quad C_{Tdes,ld}$$

These parameters are specified in the following fixed format (Fortran notation):

'CD', 2X, 5 (3X, E10.5)

Note that the  $C_{Tdes,app}$  and  $C_{Tdes,ld}$  coefficients are determined in order to obtain a 3° descent gradient during approach and landing at reference weight.

The third data line specifies the reference speeds during descent, that is:

$$V_{des,ref} \quad M_{des,ref}$$

These parameters are specified in the following fixed format (Fortran notation):

'CD', 2X, 2 (3X, E10.5)

Note that these two parameters are no longer used in performance computations, and are included only to provide information on one of the reference descent speeds used during the model identification.

The zero values at the end of this data line are not used but are included for backward compatibility reasons.

### 5.4.7. Fuel Consumption Block

The OPF fuel consumption block consists of 3 data lines with 4 comment lines for a total of 7 lines. An example of a fuel consumption block is shown below.

```

      CC===== Fuel Consumption =====/
      CC  Thrust Specific Fuel Consumption Coefficients /
1 ->  CD  .63936E+00 .10047E+04 /
      CC  Descent Fuel Flow Coefficients /
2 ->  CD  .21196E+02 .67071E+05 /
      CC  Cruise Corr.      unused      unused      unused      unused /
3 ->  CD  .98852E+00 .00000E+00 .00000E+00 .00000E+00 .00000E+00 /

```

The first data line specifies the following BADA parameters for thrust specific fuel consumption:

$C_{f1}$              $C_{f2}$

These parameters are specified in the following fixed format (Fortran notation):

'CD', 2X, 2 (3X, E10.5)

The second data line specifies the following BADA parameters for descent fuel flow:

$C_{f3}$              $C_{f4}$

These parameters are specified in the following fixed format (Fortran notation):

'CD', 2X, 2 (3X, E10.5)

The third data line specifies the cruise fuel flow correction factor:

$C_{fcr}$

The parameter is specified in the following fixed format (Fortran notation):

'CD', 5X, E10.5

#### 5.4.8. Ground Movement Block

The OPF ground movement block consists of 1 data line with 3 comment lines for a total of 4 lines. An example of a ground movement block is shown below. The ground movement block is the last block in the OPF file and is thus followed by the end-of-file line as shown below.

```

1  CC===== Ground =====/
   CC      TOL      LDL      span      length      unused      /
   CD      .23620E+04  .15550E+04  .44840E+02  .54080E+02  .00000E+00  /
   CC===== /
   FI

```

The data line specifies the following BADA parameters for ground movements:

|     |     |      |        |
|-----|-----|------|--------|
| TOL | LDL | span | length |
|-----|-----|------|--------|

These parameters are specified in the following fixed format (Fortran notation):

'CD', 2X, 4 (3X, E10.5)

## 5.5. APF FILE FORMAT

The Airlines Procedures File (APF) is an ASCII file which, for a particular aircraft type, specifies recommended speed procedures for climb, cruise, and descent conditions. An example of an APF file for a dummy jet aircraft is shown below.

[illegible]

There are two types of lines in the APF file with the line type identified by the first two characters in the line. These line types with their associated two leading characters are listed below:

CC - comment line  
CD - data line

The last line in the file, as shown above, is also a comment line.

The comment lines are provided solely for the purpose of improving the readability of the file. All coefficients are contained within the CD lines in a fixed format.

- file identification block
- procedures specification block

### 5.5.1. File Identification Block

```
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC J2H____.APF CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC/
CC                                                                                               /
CC              AIRLINES PROCEDURES FILE                                                         /
CC                                                                                               /
CC      File_name: J2H____.APF                                                                    /
CC                                                                                               /
CC      Creation_date: Apr 30 2002                                                                /
CC                                                                                               /
CC      Modification_date: Mar 05 2009                                                            /
CC                                                                                               /
CC                                                                                               /
CC                                                                                               /
CC      LO= 087.00 to ---.--   /  AV= ---.-- to ---.--   /  HI= ---.-- to 171.70                /
CC
```

The second last comment line in the identification block specifies three mass ranges for the aircraft in tonnes: low range (LO), average range (AV) and high range (HI). The definition of these ranges is used for interpreting the information presented below in the procedures specification block. In the example given above, all three ranges are assumed equivalent.

[illegible]

The first data line specifies the company name for which the next three datalines are valid. The company can be identified by its 3- and 2-letter codes plus the company name. The dataline format is:

'CD', 2X, A3, 1X, A2, 4X, A15

For now, all APF files specify procedures for only one "default" company.

The next three data lines specify the following parameters corresponding to climb, cruise and descent:

$V_{cl,1}$   $V_{cl,2}$   $M_{cl}$   $V_{cr,1}$   $V_{cr,2}$   $M_{cr}$   $M_{des}$   $V_{des,2}$   $V_{des,1}$

Note that all Mach number values are also multiplied by a value of 100. For example, the value 79 indicated for  $M_{cl}$  above corresponds to a Mach number of 0.79.

The three lines specify parameters for the three mass ranges: Low (LO), Average (AV) and High (HI) respectively. These parameters are specified in the following fixed format (Fortran notation):

'CD', 25X, 2(I3, 1X), I2, 10X, 2(I3, 1X), I2, 2X, I2, 2(1X, I3)

Note that approach values are set to zero. These values are not used but are included for backward compatibility reasons.

Also, each line specifies an aircraft version number, engine, and operational model. The operational model is always the same as the file name. The version number may provide some additional information on the aircraft version covered by the file while the engine states which engine is used by the aircraft.

The file format is designed so that the four data lines can be repeated for the different companies which operate the aircraft and which may have different standard procedures. If data were to be provided for more than one company then the version, engine and operational model fields may be useful since different companies could operate different versions of the aircraft with different engines and thus different associated operational models.

## 5.6. PTF FILE FORMAT

The Performance Table File (PTF) is an ASCII file, which for a particular aircraft type specifies cruise, climb and descent performance at different flight levels. An example of a PTF file for a dummy jet aircraft is shown below.

```

BADA PERFORMANCE FILE                                     Nov 17 2020

AC/Type: J2H_____

Source OPF File: Sep 05 2008
Source APF file: Mar 05 2009

Speeds: CAS (LO/HI) Mach Mass Levels [kg] Temperature: ISA
climb - 250/310 0.79 low - 104400
cruise - 250/310 0.79 nominal - 140000 Max Alt. [ft]: 41000
descent - 250/290 0.79 high - 171700
=====
FL | TAS CRUISE fuel | TAS CLIMB fuel | TAS DESCENT fuel
   | [kts] lo [kg/min] hi | [kts] lo [fpm] hi | [kts] ROCD [fpm] [kg/min]
   | | | | | | | | | | | | | | | | | | | | | | | | | | | |
0 | 157 2454 1925 1556 219.7 | 131 698 84.1
5 | 158 2437 1907 1536 217.8 | 132 714 83.3
10 | 159 2420 1889 1517 215.9 | 138 730 82.9
15 | 166 2530 1974 1588 214.9 | 149 774 82.9
20 | 167 2512 1955 1568 213.0 | 181 988 28.3
30 | 230 53.3 69.9 88.8 | 190 2940 2289 1852 212.9 | 230 1287 20.2
40 | 233 53.4 70.1 89.0 | 225 3474 2695 2191 214.6 | 233 1306 19.9
60 | 272 60.0 73.3 88.5 | 272 4081 2973 2285 213.7 | 272 1520 19.3
80 | 280 60.3 73.8 89.1 | 280 3932 2846 2168 206.0 | 280 1561 18.7
100 | 289 60.5 74.2 89.7 | 357 3897 2879 2256 208.7 | 334 1984 18.0
120 | 297 60.9 74.6 90.3 | 367 3687 2706 2101 200.8 | 344 2027 17.4
140 | 378 82.2 91.8 102.8 | 378 3472 2527 1941 193.0 | 354 2071 16.8
160 | 389 82.4 92.3 103.4 | 389 3250 2344 1776 185.2 | 365 2075 16.1
180 | 401 82.7 92.7 104.0 | 401 3023 2156 1607 177.4 | 376 2119 15.5
200 | 413 82.9 93.1 104.6 | 413 2790 1962 1434 169.6 | 387 2163 14.9
220 | 425 83.2 93.5 105.2 | 425 2551 1765 1256 161.8 | 399 2206 14.2
240 | 438 83.4 93.9 105.8 | 438 2308 1563 1074 154.1 | 412 2248 13.6
260 | 452 83.6 94.3 106.5 | 452 2059 1357 889 146.3 | 425 2289 13.0
280 | 466 83.8 94.7 107.1 | 466 1807 1147 700 138.6 | 438 2330 12.3
290 | 468 82.3 93.6 106.4 | 468 2417 1499 872 134.2 | 445 2349 12.0
310 | 464 77.5 89.8 103.8 | 464 2192 1359 648 124.9 | 459 2388 11.4
330 | 459 73.3 86.8 102.1 | 459 2214 1111 405 115.8 | 459 3297 10.8
350 | 455 69.7 84.4 101.2 | 455 1919 842 142 106.8 | 455 3198 10.1
370 | 453 66.8 83.0 101.4 | 453 1477 511 0 98.1 | 453 2882 9.5
390 | 453 64.6 82.4 102.7 | 453 1180 229 0 89.7 | 453 2873 8.9
410 | 453 62.9 82.6 104.9 | 453 859 0 0 81.5 | 453 2892 8.2
=====

```

The OPF and APF files are generated as a result of a modelling process using MatLab [RD4]. Once these two files are generated, the PTF can be automatically generated. A brief summary of the format of these files is given below.

The header of each PTF file contains information as described below.

file creation date: This is in the first line, at the top-right corner

aircraft type: This is in the third line.

source file dates: The last modification dates of the OPF and APF files which were used to create the PTF file are given in the 4th and 5th lines respectively.

Speeds: The speed laws for climb, cruise and descent are specified in lines 8, 9 and 10, that is:

|         |                                      |           |
|---------|--------------------------------------|-----------|
| climb   | $\min(V_{cl,1}, 250kt) / V_{cl,2}$   | $M_{cl}$  |
| cruise  | $\min(V_{cr,1}, 250kt) / V_{cr,2}$   | $M_{cr}$  |
| descent | $\min(V_{des,1}, 250kt) / V_{des,2}$ | $M_{des}$ |

Mass levels: The performance tables provide data for three different mass levels in lines 8, 9 and 10, that is:

|         |                |
|---------|----------------|
| low     | $1.2 m_{min.}$ |
| nominal | $m_{ref}$      |
| high    | $m_{max}$      |

Note that the low mass is not the minimum mass but 1.2 times the minimum mass.

Temperature: The temperature is mentioned in line 7. All PTF files currently provide data for ISA conditions only.

Maximum altitude: The maximum altitude as specified in the OPF file,  $h_{MO}$ , is given in line 9.

The table of performance data within the file consists of 13 columns. Each of these columns is described below:

|           |                                                          |
|-----------|----------------------------------------------------------|
| Column 1  | FL                                                       |
| Column 2  | cruise TAS (nominal mass) [knots]                        |
| Column 3  | cruise fuel consumption (low mass) [kg/min]              |
| Column 4  | cruise fuel consumption (nominal mass) [kg/min]          |
| Column 5  | cruise fuel consumption (high mass) [kg/min]             |
| Column 6  | climb TAS (nominal mass) [knots]                         |
| Column 7  | rate of climb with reduced power (low mass) [ft/min]     |
| Column 8  | rate of climb with reduced power (nominal mass) [ft/min] |
| Column 9  | rate of climb with reduced power (high mass) [ft/min]    |
| Column 10 | climb fuel consumption (nominal mass) [kg/min]           |
| Column 11 | descent TAS (nominal mass) [knots]                       |
| Column 12 | rate of descent (nominal mass) [ft/min]                  |
| Column 13 | descent fuel consumption (nominal mass) [kg/min]         |

The format for data presented in each line of the table is as follows (Fortran notation):

I3, 4X, I3, 2X, 3(1X, F5.1), 5X, I3, 2X, 3(1X, I5), 3X, F5.1, 5X, I3, 2X, I5, 2X, F5.1

Further explanatory notes on the data presented in the performance tables are given below:

- (a) Cruise data are only specified for flight levels greater than or equal to 30.
- (b) Performance data are specified up to the aircraft maximum altitude.
- (c) True airspeed for climb, cruise and descent is determined based on the speed schedules specified in Sections 3.1, 3.2 and 3.3 respectively.
- (d) Rates of climb are calculated at each flight level assuming the energy share factors associated with constant CAS or constant Mach speed laws and using the reduced power correction as given in Section 2.8.
- (e) The fuel consumption in climb is independent of the aircraft mass and thus only one value is given. There are three different climb rates however corresponding to low, nominal and high mass conditions.
- (f) The rate of descent and fuel consumption in descent is calculated assuming the nominal mass. Values for other mass conditions are not provided.
- (g) Discontinuities in climb rate can occur for the following reasons:
  - change in speed between flight levels (e.g. removal of 250 kt restriction above FL100),
  - transition from constant CAS to constant Mach (typically around FL300),
  - transition through the tropopause (FL360),
  - end of the application scope for reduced climb power (at 80% of  $h_{max}$ ).
- (h) Discontinuities in descent rate can occur for the following reasons:
  - transition through the tropopause (FL360),
  - transition from constant Mach to constant CAS,
  - change in assumed descent thrust (specified by the BADA  $h_{des}$  parameter),
  - change to approach or landing aerodynamic configuration,
  - change in speed between flight levels (e.g. application of 250 kt limit below FL100).
- (i) The PTF files are generated using "non-clean" configuration data for approach and landing when such data are available (see Section 2.6.1).
- (j) The performance data presented in the table are computed by using 'point type' calculation, that is without performing integration over time: aircraft weight is constant and does not account for consumed fuel, and speed changes take place immediately.
- (k) The flight envelope limitations are not taken into account for calculation of performance parameters<sup>10</sup>.

---

<sup>10</sup> Example: cruise fuel flow is calculated without checking, for given aircraft weight, speed and FL, that aircraft drag is lower than maximum available cruise thrust.

## 5.7. PTD FILE FORMAT

In addition to the data provided in the PTF file, more detailed climb and descent performance data are presented in the PTD file. An example of a PTD file for a dummy jet aircraft is shown below (partial listing):

| Low mass CLIMBS |      |        |            |        |         |         |      |          |           |         |           |        |          |        |        |
|-----------------|------|--------|------------|--------|---------|---------|------|----------|-----------|---------|-----------|--------|----------|--------|--------|
| =====           |      |        |            |        |         |         |      |          |           |         |           |        |          |        |        |
| FL[-]           | T[K] | p[Pa]  | rho[kg/m3] | a[m/s] | TAS[kt] | CAS[kt] | M[-] | mass[kg] | Thrust[N] | Drag[N] | Fuel[kgm] | ESF[-] | ROC[fpm] | TDC[N] | PWC[-] |
| 0               | 288  | 101325 | 1.225      | 340    | 136.35  | 136.35  | 0.21 | 104400   | 297160    | 85670   | 215.8     | 0.98   | 2452     | 186284 | 0.88   |
| 5               | 287  | 99508  | 1.207      | 340    | 137.34  | 136.35  | 0.21 | 104400   | 294268    | 85680   | 213.9     | 0.98   | 2435     | 183727 | 0.88   |
| 10              | 286  | 97717  | 1.190      | 339    | 138.34  | 136.35  | 0.21 | 104400   | 291385    | 85691   | 212.0     | 0.98   | 2417     | 181179 | 0.88   |
| 15              | 285  | 95952  | 1.172      | 339    | 144.45  | 141.35  | 0.22 | 104400   | 288510    | 82072   | 211.0     | 0.97   | 2527     | 181833 | 0.88   |
| 20              | 284  | 94213  | 1.155      | 338    | 145.51  | 141.35  | 0.22 | 104400   | 285643    | 82082   | 209.1     | 0.97   | 2509     | 179299 | 0.88   |
| 30              | 282  | 90812  | 1.121      | 337    | 168.52  | 161.35  | 0.26 | 104400   | 279935    | 72295   | 209.0     | 0.96   | 2937     | 182892 | 0.88   |
| 40              | 280  | 87511  | 1.088      | 336    | 202.72  | 191.35  | 0.31 | 104400   | 274260    | 67093   | 210.7     | 0.95   | 3470     | 182476 | 0.88   |
| 60              | 276  | 81200  | 1.024      | 333    | 272.30  | 250.00  | 0.42 | 104400   | 263011    | 74643   | 213.7     | 0.91   | 4075     | 165917 | 0.88   |
| 80              | 272  | 75262  | 0.963      | 331    | 280.34  | 250.00  | 0.44 | 104400   | 251895    | 74535   | 206.0     | 0.91   | 3925     | 156222 | 0.88   |
| 100             | 268  | 69682  | 0.905      | 328    | 345.37  | 300.00  | 0.54 | 104400   | 240914    | 91120   | 207.0     | 0.87   | 3905     | 131941 | 0.88   |
| 120             | 264  | 64441  | 0.849      | 326    | 355.51  | 300.00  | 0.56 | 104400   | 230066    | 90785   | 199.1     | 0.86   | 3703     | 122681 | 0.88   |
| 140             | 260  | 59524  | 0.796      | 324    | 366.04  | 300.00  | 0.58 | 104400   | 219352    | 90425   | 191.3     | 0.85   | 3495     | 113561 | 0.88   |
| 160             | 256  | 54915  | 0.746      | 321    | 376.97  | 300.00  | 0.60 | 104400   | 208772    | 90038   | 183.6     | 0.84   | 3281     | 104583 | 0.88   |
| 180             | 252  | 50600  | 0.698      | 319    | 388.32  | 300.00  | 0.63 | 104400   | 198326    | 89622   | 175.8     | 0.83   | 3060     | 95748  | 0.88   |
| 200             | 249  | 46563  | 0.653      | 316    | 400.10  | 300.00  | 0.65 | 104400   | 188013    | 89175   | 168.1     | 0.82   | 2834     | 87059  | 0.88   |
| 220             | 245  | 42791  | 0.610      | 314    | 412.32  | 300.00  | 0.68 | 104400   | 177835    | 88694   | 160.4     | 0.81   | 2601     | 78516  | 0.88   |
| 240             | 241  | 39271  | 0.569      | 311    | 425.00  | 300.00  | 0.70 | 104400   | 167790    | 88179   | 152.7     | 0.80   | 2364     | 70123  | 0.88   |
| 260             | 237  | 35989  | 0.530      | 308    | 438.16  | 300.00  | 0.73 | 104400   | 157879    | 87627   | 145.0     | 0.79   | 2122     | 61879  | 0.88   |
| 280             | 233  | 32932  | 0.493      | 306    | 451.80  | 300.00  | 0.76 | 104400   | 148102    | 87036   | 137.3     | 0.78   | 1875     | 53788  | 0.88   |
| 290             | 231  | 31485  | 0.475      | 304    | 458.81  | 300.00  | 0.78 | 104400   | 143263    | 86725   | 133.4     | 0.78   | 1749     | 49800  | 0.88   |
| 310             | 227  | 28745  | 0.442      | 302    | 463.54  | 293.28  | 0.79 | 104400   | 133687    | 83916   | 124.9     | 1.09   | 2184     | 43839  | 0.88   |
| 330             | 223  | 26201  | 0.410      | 299    | 459.48  | 280.58  | 0.79 | 104400   | 124245    | 79587   | 115.8     | 1.09   | 2205     | 44658  | 1.00   |
| 350             | 219  | 23842  | 0.380      | 297    | 455.37  | 268.17  | 0.79 | 104400   | 114936    | 75881   | 106.8     | 1.09   | 1911     | 39055  | 1.00   |
| 370             | 217  | 21663  | 0.348      | 295    | 453.12  | 256.08  | 0.79 | 104400   | 105761    | 72808   | 98.1      | 1.00   | 1470     | 32953  | 1.00   |
| 390             | 217  | 19677  | 0.316      | 295    | 453.12  | 244.46  | 0.79 | 104400   | 96720     | 70398   | 89.7      | 1.00   | 1174     | 26322  | 1.00   |
| 410             | 217  | 17874  | 0.287      | 295    | 453.12  | 233.34  | 0.79 | 104400   | 87813     | 68640   | 81.5      | 1.00   | 855      | 19173  | 1.00   |

The performance values presented in the PTD file are a superset of the climb and descent performance values presented in the PTF file. They are generated in the same conditions as the corresponding PTF file: same aircraft, same source OPF and APF files, same speed laws, same mass levels, same temperature and same flight levels. The purpose of this file is mainly to provide the user with a greater number of computed parameters, especially intermediate parameters used to compute the final TAS and ROCD, which may be useful to validate an implementation of the BADA model.

The files contain performance data consisting of 4 sections:

- low mass climb performance
- nominal mass climb performance
- high mass climb performance
- nominal mass descent performance

Each section contains a table that presents, for several flight levels, a set of performance parameters spread across 16 columns. Each of these columns is described below:

|           |                                                                                                                                |
|-----------|--------------------------------------------------------------------------------------------------------------------------------|
| Column 1  | Flight level [FL]                                                                                                              |
| Column 2  | Temperature [K]                                                                                                                |
| Column 3  | Pressure [Pa]                                                                                                                  |
| Column 4  | Air density [kg/m <sup>3</sup> ]                                                                                               |
| Column 5  | Speed of sound [m/s]                                                                                                           |
| Column 6  | TAS [kt]                                                                                                                       |
| Column 7  | CAS [kt]                                                                                                                       |
| Column 8  | Mach [dimensionless]                                                                                                           |
| Column 9  | Mass [kg]                                                                                                                      |
| Column 10 | Thrust [N]                                                                                                                     |
| Column 11 | Drag [N]                                                                                                                       |
| Column 12 | Fuel flow [kg/min]                                                                                                             |
| Column 13 | Energy share factor [dimensionless]                                                                                            |
| Column 14 | Rate of climb/descent [ft/min]                                                                                                 |
| Column 15 | (Thr – D) · C <sub>pow,red</sub> [kg/min] (see section 3.8)                                                                    |
| Column 16 | - climb tables: Power reduction coefficient C <sub>pow,red</sub> [dimensionless]<br>- descent table: Descent gradient [degree] |

The format for data presented in each line of the table is as follows (Fortran notation):

Climb tables:

I6, 1X, I3, 1X, I6, 1X, F7.3, 1X, I7, 2(1X, F8.2), 1X, F7.2, 1X, I6, 2(1X, I9), 1X, F7.1, 1X, F7.2, 1X, I7, 1X, I8, 1X, F7.2

Descent tables:

I6, 1X, I3, 1X, I6, 1X, F7.3, 1X, I7, 2(1X, F8.2), 1X, F7.2, 1X, I6, 2(1X, I9), 1X, F7.1, 1X, F7.2, 1X, I7, 1X, I8, 1X, F8.2

## 5.8. BADA.GPF FILE FORMAT

The BADA.GPF file is an ASCII file that specifies the values of the global aircraft parameters (see Section 4). The complete BADA.GPF file is shown below:

[illegible]

```

CD V_cl_5          mil,civ jet          cl          .80000E+02 /
CC spd incr FL < 5 [KCAS]                /
CD V_cl_6          mil,civ tbp,pst,elc   cl          .20000E+02 /
CC spd incr FL < 10 [KCAS]                /
CD V_cl_7          mil,civ tbp,pst,elc   cl          .30000E+02 /
CC spd incr FL < 15 [KCAS]                /
CD V_cl_8          mil,civ tbp,pst,elc   cl          .35000E+02 /
CC spd incr FL < 10 [KCAS]                /
CD V_des_1         mil,civ jet,tbp       des         .50000E+01 /
CC spd incr FL < 15 [KCAS]                /
CD V_des_2         mil,civ jet,tbp       des         .10000E+02 /
CC spd incr FL < 20 [KCAS]                /
CD V_des_3         mil,civ jet,tbp       des         .20000E+02 /
CC spd incr FL < 30 [KCAS]                /
CD V_des_4         mil,civ jet,tbp       des         .50000E+02 /
CC spd incr FL < 5 [KCAS]                /
CD V_des_5         mil,civ pst,elc       des         .50000E+01 /
CC spd incr FL < 10 [KCAS]                /
CD V_des_6         mil,civ pst,elc       des         .10000E+02 /
CC spd incr FL < 15 [KCAS]                /
CD V_des_7         mil,civ pst,elc       des         .20000E+02 /
CC hold. spd FL < 140 [KCAS]              /
CD V_hold_1        mil,civ jet,tbp,pst,elc hold      .23000E+03 /
CC hold. spd FL < 200 [KCAS]              /
CD V_hold_2        mil,civ jet,tbp,pst,elc hold      .24000E+03 /
CC hold. spd FL < 340 [KCAS]              /
CD V_hold_3        mil,civ jet,tbp,pst,elc hold      .26500E+03 /
CC hold. spd FL > 340 [M]                  /
CD V_hold_4        mil,civ jet,tbp,pst,elc hold      .83000E+00 /
CC backtrack spd   [KCAS]                 /
CD V_backtrack     mil,civ jet,tbp,pst,elc gnd        .35000E+02 /
CC taxi spd        [KCAS]                 /
CD V_taxi          mil,civ jet,tbp,pst,elc gnd        .15000E+02 /
CC apron spd       [KCAS]                 /
CD V_apron         mil,civ jet,tbp,pst,elc gnd        .10000E+02 /
CC gate spd        [KCAS]                 /
CD V_gate          mil,civ jet,tbp,pst,elc gnd        .50000E+01 /
CC Electric pow. red. [-]                 /
CD C_red_elec      mil,civ elc            ic,cl       .00000E+00 /
CC Piston pow. red. [-]                   /
CD C_red_piston    mil,civ pst            ic,cl       .00000E+00 /
CC Turboprop pow. red. [-]                /
CD C_red_turbo     mil,civ tbp            ic,cl       .25000E+00 /
CC Jet pow. red. [-]                      /
CD C_red_jet       mil,civ jet            ic,cl       .15000E+00 /
FI=====
CC//////////////////// THE END //////////////////////////////////////

```

There are three types of lines in the BADA.GPF file with the line type identified by the first two characters in the line. These line types with their associated two leading characters are listed below.

```

CC    comment line
CD    data line
FI    end-of-file line

```

The data is organised into three blocks separated by a comment line consisting of the block name and equal signs "=":

- file identification block
- class block
- parameter block

Each of these blocks is described in the subsections below.

### 5.8.1. File Identification Block

The file identification block provides information on the file name, creation and modification date. The block consists of 12 comment lines.

```

CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC BADA.GPF CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC/
CC                                                                                        /
CC              GLOBAL PARAMETERS FILE                                                  /
CC                                                                                        /
CC      File_name: BADA.GPF                                                            /
CC                                                                                        /
CC      Creation_date: Mar 26 2002                                                      /
CC                                                                                        /
CC      Modification_date: Mar 26 2002                                                  /

```

The comment lines specify the file name along with the creation date and last modification date. The creation date indicates the date when the file was created for the first time. The modification date indicates when the contents of the file were last modified.

### 5.8.2. Class Block

The class block consists of 6 comment lines and defines the three classes (Flight, Engine and Phase) and their instances that are used in the BADA.GPF file.

```
CC===== Class =====/
CC /
CC Flight = civ,mil /
CC Engine = jet,tbp,pst,elc /
CC Phase = to,ic,cl,cr,des,hold,app,lnd,gnd /
CC /
```

|       |      |   |                  |
|-------|------|---|------------------|
| With: | civ  | = | civil flight     |
|       | mil  | = | military flight  |
|       | jet  | = | jet engine       |
|       | tbp  | = | turboprop engine |
|       | pst  | = | piston engine    |
|       | elc  | = | electric engine  |
|       | to   | = | take-off         |
|       | ic   | = | initial climb    |
|       | cl   | = | climb            |
|       | cr   | = | cruise           |
|       | des  | = | descent          |
|       | hold | = | holding          |
|       | app  | = | approach         |
|       | lnd  | = | landing          |
|       | gnd  | = | ground           |

### 5.8.3. Parameter Block

The parameter block contains the values of the global aircraft parameters. This block has 5 comment lines plus a comment line and a dataline for each parameter.

```

CC===== Parameters List =====/
CC /
CC Name Unit /
CC Parameter Flight Engine Phase Value /
CC /
1 -> CC max. long. acc. [fps2] /
CD acc_long_max civ jet,tlp,pst,elc to,ic,cl,cr,des,hold,app,lnd .20000E+01 /
2 -> CC max. norm. acc. [fps2] /
CD acc_norm_max civ jet,tlp,pst,elc to,ic,cl,cr,des,hold,app,lnd .50000E+01 /
3 -> CC nom. bank angle [deg] /
CD ang_bank_nom civ jet,tlp,pst,elc to,lnd .15000E+02 /

```

The parameter comment line contains the parameter name and its unit.

The parameter data line contains five fields:

- Parameter Field:** This field identifies the parameter.
- Flight Field:** This field identifies whether the parameter is valid for a civil flight, a military flight or both.
- Engine Field:** This field identifies the engine type (jet, turboprop, piston or electric) for which the parameter is valid.
- Phase Field:** This field identifies for which flight phase the parameter is valid. 9 different flight phases are currently defined.
- Value Field:** The value field gives the value of the parameter.

The fields above are specified in the following fixed format (Fortran notation):

'CD', 1X, A15, 1X, A7, 1X, A16, 1X, A29, 1X, E10.5

The parameter list continues until 'FI' (end of file) is reached.

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## **APPENDIX A**

### **SOLUTIONS FOR BUFFETING LIMIT ALGORITHM**

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A general solution for finding the roots of a cubic expression can be found in [RD7]. If we take expression 2.6-6, we can rewrite it as:

$$M^3 - \frac{C_{Lbo(M=0)}}{k} \cdot M^2 + \frac{\frac{W}{S}}{0.583 \cdot P \cdot k} = 0$$

Let:

$$a_1 = -\frac{C_{Lbo(M=0)}}{k}$$

$$a_2 = 0$$

$$a_3 = \frac{\frac{W}{S}}{0.583 \cdot P \cdot k}$$

Now let:

$$Q = \frac{(3 \cdot a_2 - a_1^2)}{9}$$

and:

$$R = \frac{(9 \cdot a_1 \cdot a_2 - 27 \cdot a_3 - 2 \cdot a_1^3)}{54}$$

The discriminant D is equal to:  $Q^3 + R^2$ . In our case D is usually strictly negative, which means that all roots are unequal and real. A simplified computation method with the help of trigonometry is given below:

$$X_1 = 2 \cdot \sqrt{-Q} \cdot \cos\left(\frac{\theta}{3}\right) - \frac{a_1}{3}$$

$$X_2 = 2 \cdot \sqrt{-Q} \cdot \cos\left(\frac{\theta}{3} + 120^\circ\right) - \frac{a_1}{3}$$

$$X_3 = 2 \cdot \sqrt{-Q} \cdot \cos\left(\frac{\theta}{3} + 240^\circ\right) - \frac{a_1}{3}$$

$$\text{With: } \cos \theta = \frac{R}{\sqrt{-Q^3}}$$

The solutions  $X_1$ ,  $X_2$  and  $X_3$  now give the possible values of M. One solution (in our case usually  $X_1$ ) is always negative. The others are positive with the lower one (usually  $X_2$ ) being the low-speed buffeting limit we are looking for.



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